■ প্রশ্ন কমনের প্রমাণ পত্র ■

প্রমাণপত্র : ২০২৪ সালের পুরাতন সাজেশন্স থেকে কমনকৃত প্রশ্নের হার ৯৯%।

National University

[BBA Hon's 3rd Year; Exam-2024 (Held on-29/01/2026)]

(Department of Management)

Subject Code : 232603

Subject Name : Business Statistics (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Different parts of each question must be written consecutively.]

Part-A

1. Answer any **ten** questions : $1 \times 10 = 10$

(a) What is an independent variable? *[অধ্যায়-০৭ এর ১০ নং প্রশ্ন কমন পড়েছে]*

(b) What is secondary data? *[অধ্যায়-০১ এর ১৪ নং প্রশ্ন কমন পড়েছে]*

(c) What is Struges rule? *[অধ্যায়-০১ এর ০৫ নং প্রশ্ন কমন পড়েছে]*

(d) What is standard deviation? *[অধ্যায়-০৩ এর ১০ নং প্রশ্ন কমন পড়েছে]*

(e) Elaborate ANOVA. *[অধ্যায়-০৮ এর ০২ নং প্রশ্ন কমন পড়েছে]*

(f) What is correlation? *[অধ্যায়-০৭ এর ০৮ নং প্রশ্ন কমন পড়েছে]*

(g) What is TRT? *[অধ্যায়-০৯ এর ১২ নং প্রশ্ন কমন পড়েছে]*

(h) What is cluster sampling? *[অধ্যায়-০৯ এর ০৮ নং প্রশ্ন কমন পড়েছে]*

(i) What is T-test? *[অধ্যায়-০৫ এর ০৯ নং প্রশ্ন কমন পড়েছে]*

(j) Define null hypothesis. *[অধ্যায়-০৬ এর ০৩ নং প্রশ্ন কমন পড়েছে]*

(k) What is time series analysis? *[অধ্যায়-০৮ এর ০১ নং প্রশ্ন কমন পড়েছে]*

(l) What is skewness? *[অধ্যায়-০৩ এর ২৩ নং প্রশ্ন কমন পড়েছে]*

Part-B

[Answer any **five** questions.] Marks : $4 \times 5 = 20$

2. Discuss the differences between statistics and statistic with example. *[অধ্যায়-০১ এর ০৬ নং প্রশ্ন কমন পড়েছে]*
3. What are the methods of collecting secondary data? *[অধ্যায়-০১ এর ০৫ নং প্রশ্ন কমন পড়েছে]*
4. *[অধ্যায়-০১ এর Problem-01 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
5. *[অধ্যায়-০২ এর Problem-04 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
6. *[অধ্যায়-০৩ এর Problem-03 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
7. *[অধ্যায়-০৭ এর Problem-06 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
8. *[অধ্যায়-০৪ এর Problem-02 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
9. *[অধ্যায়-০৩ এর Problem-20 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*

Part-C

[Answer any **five** questions.] Marks : $10 \times 5 = 50$

10. (a) Discuss the importance of secondary data in statistics. *[অধ্যায়-০১ এর ০৬ নং প্রশ্ন কমন পড়েছে]*

(b) What are the differences between qualitative and quantitative data? *[অধ্যায়-০১ এর ০৭ নং প্রশ্ন কমন পড়েছে]*

11. *[অধ্যায়-০২ এর Problem-07 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
12. *[অধ্যায়-০৩ এর Problem-13 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
13. *[অধ্যায়-০৩ এর Problem-25 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
14. *[অধ্যায়-০৭ এর Problem-26 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
15. *[অধ্যায়-০৫ এর Problem-03 নং (হুবহু) প্রশ্ন কমন পড়েছে]*
16. *[অধ্যায়-০৮ এর Problem-07 নং (অনুরূপ) প্রশ্ন কমন পড়েছে]*
17. *[অধ্যায়-০৯ এর Problem-02 নং (হুবহু) প্রশ্ন কমন পড়েছে]*

National University

[BBA (Hon's) 3rd Year; Exam-2025 (Will be held-2026/27]

Subject Code : 232603

Subject Name : Business Statistics (In English)

[Baticrom Final Suggestions]

ব্যবস্থাপনা বিভাগের ছাত্রছাত্রীদের জন্য

Chapter 01 Introduction to Business Statistics

(ব্যবসায় পরিসংখ্যানের পরিচিতি)

Part-A : Brief Questions & Solutions :

✪✪ 01. **What is statistics? (পরিসংখ্যান কি?)**

[Test Exam-Tongi, Commerce, B.M. Edward, City, College; N.U. BBA (Hon's)(2nd)-2018 (Finance); 2018 (Marketing); Acc 3rd-2019]

Ans. Statistics is the science of decision making in the field of uncertainty.

02 **Who is considered as the father of modern statistics?** 99%

[N.U. BBA (Hon's) 2014; (2nd)-2018 (Acc); 2023 (Mgt. 3rd); D.U.) (2nd)-2017 (Acc]

Or, Who is the father of modern statistics?

Ans : The father of modern statistics is R.A. Fisher.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 03. **Define statistical table.**

[N.U. BBA (Hon's) (Acc)- 2nd 2016]

Ans : A statistical table means that a way of presenting statistical data through a systematic arrangement of the numbers describing some mass phenomenon or process.

04 **What is business statistics?** 99%

(ব্যবসায় পরিসংখ্যান কি?)

[N.U. (Old) 2011, 2013; 2021 (Mgt. 3rd)]

Or, What is meant by business statistics?

Or; Define business statistics.

Ans. Business statistics is a science assisting us to make decision under uncertainties based on some numerical and measurable scales.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 05. **What is data? (উপাত্ত কি?)**

[N.U. BBA (Hon's) (3rd)-2018 (Mgt.); D.U. (3rd)-2016 (Mgt.)]

Ans. Data are the collection of raw facts and figures from any sets of enquiry for the purpose of statistical analysis.

✪✪✪ 06. **What is data array?**

[N.U.(Hon's) (3rd)-2017(Mgt.)]

Ans. Data array means the arrangement of raw data by either descending or ascending order.

✪✪ 07. **What are sources of data? (তথ্য বা উপাত্তের উৎস কি?)** *[Test Exam-M.M. (Mgt College)]*

Ans : Data can be collected from three sources :

(i) Primary source;

(ii) Secondary source;

(iii) Internal Source.

08 **What is primary data?** 99%

[D.U. BBA (Hon's) (2nd) 2017 (Acc); N.U. BBA (Hon's) 2018 (Marketing); 2020, 2022 (Mgt.)]

Ans : Primary data that have been collected from the origin of the fields is called primary data.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 09. **Write the definition of continuous variable.** *[N.U BBA (Hon's) (Old) 2012]*

Ans : A continuous variable is one that can take on any value within given range.

✪✪✪ 10. **Define discrete variable.**

[D.U. BBA (Hon's) (2nd)-2017; N.U. (2nd)-2018 ((Marketing); D.U. (3rd)-2016 (Mgt.)]

Ans : A discrete variable is a variable that can only take on a certain number of values.

✪✪✪ 11. **What is ogive? (অজিভ কি?)** 99%

[Test Exam-B.L (Mgt.) College; N.U. BBA (Hon's) (3rd)-2017 (Mgt.); Acc-2019]

Ans : Ogive is a graph of cumulative frequency distribution.

✪✪ 12. **What is frequency density?** *[N.U. (Acc-2nd)-2017]*

Ans : The ratio of the frequency of a class to the class interval is called frequency density of that class.

✪✪✪ 13. **What is frequency?** *[D.U. (Marketing)- 2nd 2017]*

Ans : The repeated times of value of the variable is called frequency of that value.

14. ***Define secondary data. [২০২৪ সালে যা এসেছিল]*** *[N.U. BBA (Hon's) (2nd)-2018; 2024 (Mgt.)]*

Or, What do you mean by secondary data?

Ans : *When an organization on investigation uses the data that has already been collected and used by others, such data are called secondary data.*

✪✪✪ 15. **What is frequency polygon? (গণসংখ্যা বহুভুজ কি?)**

Ans : A frequency polygon expresses distribution of data by means of a single line determined by the midpoints of classes.

✪✪✪ 16. **What is questionnaire?** *[N.U. (2nd)-2018 (Finance)]*

Ans : A questionnaire is written list of question which are answered by a lot of people in order to provide information to report a survey.

✪✪✪ 17 **What is meant by histogram? (আয়তলেখ কি?)**

[N.U. 2011, 2013, (Acc)-2016; N.U. BBA (Hon's) (2nd)-2018 (Marketing); 2020 (Mgt.)]

Or, What is histogram? 99%

Ans : Histogram is a suitable graph for representing the frequency distribution of a continuous series.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 18. **What is class interval? (শ্রেণিব্যাপ্তি কি?)**

[D.U. BBA (Hon's) (Finance)- 2nd 2017]

Ans : Difference between the lower limit class and upper limit class is called class interval.

✪✪ 19. **What is meant by classification?**

[N.U. BBA (Hon's) (Old) 2014, (Acc)- 2nd 2015]

Or, What is classification?

Ans : Classification is the process of arranging data as collected by proper investigation in some groups or class in relation with definite..

✪✪✪ 20. **What is meant by descriptive statistics?** 99%

[N.U. 2022 (Hon's) 3rd]

Ans : Descriptive statistics are the methods that help to collect, summarize, present and analyse a set of data.

✪✪ 21. **Write down the Struggles rule. (স্ট্রাইগেজের সূত্র লিখ।)**

Ans : Class interval $= \frac{\text{Range}}{1+3.322\log n}$

22. ***What is Struges rule? [২০২৪ সালে যা এসেছিল]***

[N.U. 2024 (Mgt. 3rd)]

Ans : *Struges rule is a statistical rule used to determine the number of class intervals when constructing a frequency distribution.*

✪✪✪ 23 **What is population?** 99%

[N.U. BBA (Hon's) (Old) 2014; 2021 (Mgt. 3rd)]

Ans : Population refers to all subject possessing a common characteristic that is being studied.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 24. **What is BBS?** *[N.U. BBA (Hon's) (Acc)-2016]*

Ans : The Bangladesh Bureau of statistics is the centralized official bureau is Bangladesh for collecting statistics on demography the economy and other facts about the country and disseminating the information.

✪✪✪ 25. **What is pie-chart?** 99% *[N.U. (Mgt.) 2020]*

Ans. A pie-chart is formed by marking off a portion of the pie corresponding to each characteristic being displayed.

✪✪ 26. **What is raw data? (কাঁচা উপাত্ত কি?)**

[N.U. (2nd)-2018 (Finance)]

Ans : Information before it is arranged or analyzed by statistical methods.

✪✪✪ 27. **What is frequency distribution? (গণসংখ্যা নিবেশন কি?)**

Ans : An organized display of data that shows the number of observations in each of a set of mutually exclusive and collectively exhaustive classes is called frequency distribution.

✪✪ 28. **What is unequal class interval? (অসম শ্রেণি ব্যাপ্তি কি?)**

Ans : The distribution which width of each class is not equal is called an unequal class interval.

✪✪ 29. **What is bi-variate? (দ্বিচলক কি?)**

Ans : A frequency table where two variables have been measured in the same set of items through cross classification is known as bi variate frequency distribution.

Part-B : Short Questions :

✪✪ 01. What is statistics? [উ : ব্যবস্থাপনা হ্যান্ডনোটস (২য় খণ্ড) বইয়ের ৪ পৃষ্ঠার ১.০১ নং]
Or, Definition of statistics? *[N.U. BBA (Hon's) (Old) 2004, 2006, 2009;(Finance 2nd)-2017]*
Or, What do you mean by statistics?

✪✪ 02. Describe the functions of statistics. [উ : ১৩ পৃষ্ঠার ১.৪০ নং]
Or, Write down the functions of statistics.
[Test Exam : Commerce, B.M. Edward, City College; N.U. BBA (Hon's) (Old)2008 ;(Finance 2nd)-2018]

✪✪✪ 03. **Describe the limitations of statistics.** 99% [উ : ৫ পৃষ্ঠার ১.০৬ নং]
[N.U. BBA (Hon's) 2013; (Acc 2nd) 2018; (Marketing 2nd) 2016, 2018;Mgt 2016, 2019; D.U. 2017 (Managment-3rd); (Acc 3rd)-2019]
Or, Discuss briefly the various limitations of statistics.
Or; Write down the demerits of statistics.

✪✪ 04. Discuss the types of variables. [উ : ৯ পৃষ্ঠার ১.২৩ নং] *[N.U. 2023 (Mgt.)]*

✪✪✪ 05. **Discuss the scope of business statistics.** 99% [উ : ৪ পৃষ্ঠার ১.০২ নং] *[N.U. 2021 (Mgt.)]*

✪✪✪ (06) **What is the role of statistics in managerial decision making?** 99% [Ans : Page No. 14; Ques. No 1.41]
[N.U. (Old) 2004, 2007, 2008, 2009, 2010, (Acc 2nd)-2015, 2017; (Finance 2nd) 2015; (Marketing 2nd) 2017; (3rd)-2017, 2018, 2020, 2022]
Or, Discuss the importance of statistics in business.
Or, Why is statistics required in business?
[রকেট স্পেশাল সাজেশন্স: ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

✪✪✪ (07) **Show the differences between primary and secondary data.** 99% [উ : ৬ পৃষ্ঠার ১.১২ নং]
[Test Exam : Eden, B.M, Al Hera, Satkhira, Shahid Munsur Colleg; N.U. BBA (Hon's) (Old) 2004, 2006, 2009, 2012, 2013; (Finance 2nd) 2015, 2017; (Marketing 2nd) 2017, 2020; Mgt 2020, 2023; D.U. (Managment 3rd)- 2017].
[রকেট স্পেশাল সাজেশন্স: ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল–

2. *Discuss the differences between statistics and statistic with example.* [Ans : Page No. 562 Ques. NO- 2] *[N.U. 2024 (Mgt.)]*
3. *What are the methods of collecting secondary data?* [Ans : Page No. 7 Ques. No 1.14]
Or, Describe the sources of secondary data. *[N.U. BBA (Hon's) (3rd)-2019, 2021, 2024 (Mgt.)]*

Part-C : Broad Questions :

✪✪ 01. What is business statistics? [উ : ৪ পৃষ্ঠার ১.০৩ নং]
Or, Definition of business statistics? *[N.U. BBA (Hon's) (Old) 2005, 2008; D.U. (Affiliated) (Finance 2nd) 2017]*

✪✪ 02. Show the difference between inclusive class interval and exclusive class interval. [উ : ১৩ পৃষ্ঠার ১.৩৯ নং]
[N.U. BBA (Hon's) (All) 2007, 2009; (Finance & Marketing 2nd) 2017]

✪✪✪ 03. **Write down the important characteristics of statistics.** 99% [উ : ৫ পৃষ্ঠার ১.০৭ নং]
Or, Discuss the characteristics of business statistics. *[N.U. BBA (Hon's) (All) 2012; (3rd)-2017 (Mgt.)]*

✪✪ 04. Define classification and state four primary rules that should be observed while classifying data. [উ : ১১ পৃষ্ঠার ১.৩১ নং] *[N.U. BBA (Hon's) (Acc 2nd)-2016]*

✪✪✪ (05) **What do you mean by bi-variate frequency distribution?** 99% [উ : ১৬ পৃষ্ঠার ১.৪৬ নং]
Or, What is bi-variate frequency distribution. */Test Exam : Bagbodnu, Thakurga College; N.U. 2009, (3rd)-Mgt 2019]*
[রকেট স্পেশাল সাজেশন্স: ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল–

10. (a) *Discuss the importance of secondary data in statistics.* [Ans : Page No. 566 Ques. NO- 10 (a] *[N.U. 2024 (Mgt.)]*
(b) *What are the differences between qualitative and quantitative data?* [Ans : Page No. 566 Ques. NO- 10 (b)] *[N.U. 2024 (Mgt.)]*

Help Line ***Dear Student's, In above Suggestions- 06 & 07 No. in Part-B and 03 & 05 No. Questions in Part-C are most Important for the Examinee of 2025. – Editor***

Easy Math Basic Problem : Category-B & C

☑ আয়তলেখ (Histogram)

○○ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2017 No–3**

From the following data construct a histogram:

Class interval	20 — 29	30—39	40 — 49	50—59
Frequency	20	35	15	30

Solution : ব্যবস্থাপনা সাজেশন (২য় খণ্ড) বইয়ের ২০ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল]

☒☒ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2024 No–4**

From the following data draw a histogram :

Class interval	20 – 25	25 – 30	30 – 35	35 – 40	40 – 45
Frequency	8	12	15	10	8

Solution : ৫৬২ পৃষ্ঠার **Question No-04** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2019 No– 04**

Represent the following data by histogram and calculate mode from histogram :

Yearly Income	10—20	20—30	30—40	40—50	50—60	60—70
No. of Family	125	350	170	60	25	10

Solution : ২১ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–10; D.U. BBA (Hon's) 2nd (Accounting) 2017 No–11**

The daily profits in taka of 100 shops are distributed as follows :

Draw a histogram of the data and then find the modal value. Check this value. Check direct computation :

Profits per shop	0 – 100	100 – 200	200 – 300	300 – 400	400 – 500	500 – 600
No. of shops	12	18	27	20	17	6

Solution : ২৪ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-05** ***Year Question & Solution*** ⊙ **D.U. BBA (Hon's) 2nd (Marketing) 2017 No–3**

Draw the histogram from the following data.

Midpoint (X)	105	115	125	135	145	155	165
Frequency (f)	11	28	36	49	33	20	8

Solution : ২৫ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

☑ গণসংখ্যা বহুভুজ (Frequency Polygon)

○○○ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2022 No– 03**

Draw a frequency polygon from the following data :

Class interval	Frequency
20 – 29	50
30 – 39	20
40 – 49	80
50 – 59	15
60 – 69	35

Solution : ২৮ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

☑ Ogive রেখা অথবা অজিভ রেখা

oo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2017 No–11**

From the following data draw a less than ogive and determine the number of companies who earns profit of Tk. 45 crore to Tk. 75 crore :

Profit (Tk. in crore)	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	90–100
No. of company	8	12	20	24	15	10	7	3	1

Solution : ৩১ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2022 No– 15**

Below is given the frequency distribution of weekly wages of 100 workers in a factory :

Monthly wages (Tk.)	No. of workers
3000 – 3500	3
3500 – 4000	5
4000 – 4500	12
4500 – 5000	23
5000 – 5500	31
5500 – 6000	10
6000 – 6500	8
6500 – 7000	5
7000 – 7500	3

Draw the ogive for the distribution and use it to determine the median of a worker.

Solution : ৩২ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2023 No–10**

Draw an ogive from the following table and calculate the median from ogive.

Wages per hour	10 – 15	15 – 20	20 – 25	25 – 30	30 – 35	35 – 40
No. of workers	8	11	12	16	12	7

Solution : ৩৫ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

oo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2015 No–5**

Draw less than and more than ogives from the data given below:

Marks	15 – 25	25 – 35	35 – 45	45 – 55	55 – 65	65 – 75
No. of students	5	3	7	5	3	7

Solution : ৩৬ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2018 No–10**

Draw an ogive curve from the following table and show quartiles in the graph. Also find the number of workers whose wages are above 50% :-

Wages (in Th. Tk.)	50 – 55	55 – 60	60 – 65	65 – 70	70 – 75	75 – 80
Number of workers	7	11	78	30	12	2

Solution : ৩৭ পৃষ্ঠার **Problem-15** নং প্রশ্নের উত্তর দেখুন।

☑ বৃত্ত চিত্র (Pie Chart)

○○ **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2015 No–4**

The following table shows the area in millions of square kilometres of the oceans of the world:

Ocean	Area (million sq. km)
Pacific	70.8
Atlantic	41.2
Indian	28.5
Antarctic	7.6
Arctic	4.8

Draw a pie diagram to represent the data.

Solution : ৪০ পৃষ্ঠার **Problem-17** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2014 No–3; (Marketing) 2017 No–4; 3rd (Mgt) 2016 No–3; 2021 No–5**

Draw a pie-chart for the following data:

Cost of Materials Tk. 40,000; Cost of labour Tk. 30,000; Direct Expenses of manufacture; Tk. 10,000; Factory Overhead expenses Tk. 15,000; Misc. Expenses Tk. 5,000

Solution : ৪১ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

☑ দণ্ড চিত্র (Bar Diagram)

○○ **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2018 No–4**

Draw a suitable diagram from the following data :

Expenditure	Income	
	Family - A	Family - B
Food	400	250
Clothing	100	150
Education	160	70
Saving or deficit	20	– 30

Solution : ৪২ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

☑ Stem and Leaf Diagram এর অংক

○○○ **Problem-15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–7; (Mgt) 3rd 2018 No–3; 2023 No–4**

Construct a Stem and Leaf Display from the following data :

10	8	7	5	10	13	25	21	30	16	19	40
30	50	28	29	32	33	20	26	37	39	24	42

Solution : ৪৪ পৃষ্ঠার **Problem-20** নং প্রশ্নের উত্তর দেখুন।

০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

○○○ **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2016 No–10; 2021 No–10**

The obtained marks of 30 candidates in an examination are given below :

40	30	35	55	68	72	60	52	48	62
45	47	33	57	75	87	56	49	45	39
40	45	57	59	62	65	72	73	85	90

(i) Prepare a frequency distribution table by using suitable class interval;

(ii) Construct a steam and leaf diagram.

Solution : ৪৪ পৃষ্ঠার **Problem-21** নং প্রশ্নের উত্তর দেখুন।

☑ Ungroup বা Discrete গণসংখ্যা নিবেশন ছক

○○ **Problem-17** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2016 No–4**

Suppose, 20 students of BBA (Hon's)-2nd year, obtained the following marks of business statistics out of 50 marks in a class test :

20	14	17	20	17	20	16	11	17	16
17	14	14	17	18	20	12	11	14	11

You are required to present the data in a discrete frequency distribution.

Solution : ৪৫ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

☑ Bi-Variate গণসংখ্যা নিবেশন ছক

○○○ **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2017 No–10 (Finance) 2015 No–10; (Mgt) 3rd 2019 No–16(b)**

The following data represent the marks in business mathematics (x) and business statistics (y) of 13 students. Prepare a Bi-variate frequency table from the following data.

X :	32	15	21	27	58	37	47	58	46	50	40	54	62
Y :	75	48	54	70	92	53	74	85	65	62	73	77	40

Solution : ৪৭ পৃষ্ঠার **Problem-24** নং প্রশ্নের উত্তর দেখুন।

☑ Range & Limit

○○ **Problem-19** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2016 No–3**

The profits (in lac Tk.) of 30 companies for the year 2013-2014 are given below:—

22,	20,	42,	35,	37,	42,	48,	49,	53,	65,	39,	48,	67,	18,	53
23,	16,	37,	53,	49,	36,	56,	55,	54,	58,	57,	69,	25,	29,	56

Classify the above data taking a suitable class interval.

Solution : ৪৯ পৃষ্ঠার **Problem-25** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-20** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2011 No–2(b)**

For the following data prepare a frequency distribution with a class interval of 5 on inclusive basis :-

12	14	40	16	10	10	19	20	28	30
19	27	15	21	33	45	7	19	20	26
26	37	6	5	20	30	37	17	21	20

Solution : ৫০ পৃষ্ঠার **Problem-26** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-21** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2016 No–10**

The following data represent the daily income (in Tk.) of 30 workers. By taking class interval 5, construct a frequency distribution and how much percentage of workers would have income more than or equal to Tk. 50.
65, 80, 55, 60, 70, 72, 85, 50, 52, 60, 57, 87, 75, 68, 65, 56, 78, 80, 86, 62,70,60,50,56,74,66,62,88,80,75.

Solution : ৫২ পৃষ্ঠার **Problem-28** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-22** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2017 No–11(b)**

The weekly wages of 20 employees in a firm are 130, 62, 145, 76, 118, 125, 151, 142, 110, 98, 65, 116, 100, 103, 71, 85, 80, 122, 132 and 95. The firm gives bonuses of Tk. 10, 15, 20, 25 and 30, respectively in wage sub-groups of Tk. (60–80), Tk. (81–100), Tk. (101–120), Tk. (121–140), Tk. (141–160).
Show the data in frequency distribution taking wage sub-groups as class interval and find the average bonus per employee.

Solution : ৫৩ পৃষ্ঠার **Problem-29** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-23** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2019 No–11**

The following are the weekly wages of 40 employees in a firm :

120	133	125	140	155	120	166	152	143	150
111	101	145	128	131	147	135	171	156	160
104	100	170	138	130	124	127	132	154	170
128	131	158	142	140	128	112	110	115	166

(a) Construct a frequency distribution with suitable size of class interval.

(b) Draw a Histogram and hence find the modal value.

Solution : ৫৩ পৃষ্ঠার **Problem-30** নং প্রশ্নের উত্তর দেখুন।

Chapter 02 Measures of Central Tendency

(কেন্দ্রীয় প্রবণতার পরিমাপসমূহ)

Part-A : Brief Questions & Solutions :

০০০ 01. What is central tendency. (কেন্দ্রীয় প্রবণতা কি?)

[Test Exam-Eden College;N.U. BBA (Hon's) (Old) 2011, 2014, (Acc)- 2015; (2nd)-2018 (Marketing); (3rd)-2017 (Mgt.); D.U. (2nd)-2017 (Marketing)]

Or, Define central tendency. 99%

Or, What is meant by central tendency?

Ans : Central tendency is a typical value of a data set around which other figures congregate.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

০০০ 02. What are the measures of central tendency? (কেন্দ্রীয় প্রবণতার পরিমাপসমূহ কি কি?)

Or, Central limit Theorem. *[D.U. (2nd)- 2017(Acc)]*

Ans : There are 5 measures of central tendency :

(1) Arithmetic mean;

(2) Median;

(3) Mode;

(4) Geometric mean;

(5) Harmonic mean.

০০০ 03. What is central limit theorem?

[D.U. BBA (Hon's) 2017 (Finance)]

Ans : The central limit theorem states that when samples from data set with a known variance are aggregated their mean roughly equals the population mean.

০০ 04. What is quartile? (চতুর্থক কি?)

Ans : The values that divide the variables of series into four equal parts that are called the quartile. *[N.U. BBA (Hon's) (Acc)- 2017]*

০০০ 05. Define decile. (দশমক কি?)

Ans : The value/values which divides the series into ten equal parts are called deciles.

০০০ 06. What is percentile? (শতমক কি?)

Ans : The Percentiles of a set of data are those values which divides the total observations into 100 equal parts.

০০ 07. When mode is ill defined?

Ans : If a series of observations has more than one mode, then the mode is said to be ill defined mode.

০০০ 08. Define geometric mean. (জ্যামিতিক গড় কি?)

Ans : Geometric mean is the root of the product of nth variables.

০০ 09. What is harmonic mean? (তরঙ্গ গড় কি?)

Ans : Harmonic mean is the ratio between the number of items and the sum of reciprocals of items.

০০০ 10. Mention the name of the scholar who has devised the stem and leaf display. 99%

[N.U. BBA (Hon's) 2023 (Mgt.)]

Ans : The stem and leaf display was devised by the American statistician John Wilder Turkey.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

০০০ 11. What is average? (গড় কি?)

[N.U. BBA (Hon's) (Acc.); (Finance)- 2nd 2018]

Ans : Average is an attempt to find one single figure to describe whole of figures.

০০০ 12. What is arithmetic mean? (গাণিতিক গড় কি?) 99% *[N.U. BBA (Hon's) 2020 (Mgt.)]*

Ans : An arithmetic mean is a numerical value obtained by dividing the sum of values of all the observations by the total number of observations.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

✪✪ 13. **What do you understand by "Weighted mean"?**
[N.U. BBA (Hon's) (Acc)-2016; D.U. (Finance)-2017]
Or, Weighted Mean.
Ans : Weighted means a mean where some values contribute more than others.

✪✪✪ 14. **Write down the uses of Harmonic Mean. (তরঙ্গ গড়ের ব্যবহার লিখ।)**
Ans : The harmonic mean is particularly useful for computation of average rates and ratios.

✪✪ 15. **If Mean > Median > Mode, what does it indicate?**
Ans : Positively skewed distribution.

✪✪✪ 16. **The mean of a symmetrical distribution is 40.75, find its median and mode.** 99%
[N.U. 2023 (Mgt.)]
Ans : In a symmetrical distribution the mean, median and mode are all equal because the data is evenly distributed around the center. So, its median and mode are 40.75.

✪✪✪ 17. **Which measure is the best average? (কোন পরিমাপটি সর্বোৎকৃষ্ট?)**
Ans : Arithmetic mean is the best average.

✪✪✪ 18. **What is relationship among Arithmetic mean, Geometric mean and Harmonic mean? (গাণিতিক গড়, জ্যামিতিক গড় ও তরঙ্গ গড়ের সম্পর্ক কি?)**
Ans : $AM \geq GM \geq HM$

✪✪✪ 19. **What is stem and leaf display? (ফান্ড ও পত্র কি?)** 99%
[N.U. (Old) 2012, (Acc)-2017; D.U. (3rd)-2016 (Mgt.)]
Ans : One technique used to display quantitative information in a condensed form is the stem and leaf display.

✪✪✪ 20. **What is bar diagram?** *[N.U. 2018 (Acc)]*
Ans : Bar diagrams are the most common type of diagrams used in practice. A bar is a thick line whose width is shown merely for attention.

✪✪✪ 21 **What is mode? (প্রচুরক কি?)** 99%
[N.U. 2022 (Mgt. 3rd)]
Ans : Mode is the value which occurs the greatest number of frequency in a series. Mode is the most fashionable or typical value of a distribution, because it is repeated the highest number of times in the series.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 22. **What is modal class?**
(প্রচুরক শ্রেণি কি?) *[N.U. 2019 (Acc 3rd)]*
Ans : The class which belongs highest frequency is called modal class.

✪✪✪ 23. **Find the Geometric mean of the series 2, 4 and 8.** 99% *[N.U. 2022 (Mgt. 3rd)]*
Ans : Let, $x_1 = 2, x_2 = 4, x_3 = 8$

$$\therefore \text{Geometric mean} = (x_1 . x_2 . x_3)^{\frac{1}{3}}$$
$$= (2 \times 4 \times 8)^{\frac{1}{3}}$$
$$= (64)^{\frac{1}{3}}$$
$$= (4^3)^{\frac{1}{3}} = 4$$

✪✪✪ 24. **What is the mode of 15, 18, 15, 20, 25 and 15?**
[N.U. 2018(Acc)]
Ans : Given that, 15, 18, 15, 20, 25 and 15
Since 15 is the maximum number of times (i.e: 3 times) Therefore the mode is 15

✪✪✪ 25 **When AM, GM and HM are equal?** 99%
[N.U. 2021 (Mgt. 3rd)]
Ans : If each value of a observation are equal then arithmetic mean, geometric mean and harmonic are equal.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪ 26. **What is the mode of 4, 6, 8, 10, 6 and 12?**
Ans : 6. *[N.U. (Old) 2013]*

✪✪✪ 27. **Find the harmonic mean of the series 3, 6, 9.**
[N.U. (2nd)-2018 (Marketing)]
Ans : Harmonic Mean 4.91

✪✪✪ 28. **What is median?** *[N.U. 2016 (Mgt. 3rd year)]*
And : The median is the measure of central tendency which takes place in middle of an ordered sequence of values..

Part-B & C : Short & Broad Questions :

✪✪ 01. When geometric mean is preferred to arithmetic mean. *[N.U. (Old) 2005, 2008]*
[উ ব্যবস্থাপনা গণিতের (২য় খণ্ড) বইয়ের ৬২ পৃষ্ঠার ২.১৯ নং]

✪✪ 02. Describe various measures of central tendency. [উ : ৬১ পৃষ্ঠার ২.১৫ নং]

✪✪✪ 03. What are the requisites of a measure of central tendency? [উ : ৬০ পৃষ্ঠার ২.১২ নং]
[Test Exam : Edward, M. M. Aziaul, Tongi, Shahid Munsur, Satkhira Colleg; N.U. (Old) 2005, 2011; D.U. (3rd)-2016 (Mgt.)]
Or, State the necessary characteristics of a satisfactory measure of central tendency.

✪✪✪ 04. **Difference between simple and weighted arithmetic mean.** 99% [উ : ৬০ পৃষ্ঠার ২.১১ নং] *[N.U. (Acc 2nd)-2017]*

ooo (05) **What are the characteristics of a good central value/average?** 99% [উ : ৬৩ পৃষ্ঠার ২.২৩ নং]
Or, Discuss the characteristics of an ideal measures of central tendency. *[N.U. (3rd)-2017 (Mgt.)]*
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo 06. Which measure of central tendency is the best? [উ : ৫৮ পৃষ্ঠার ২.০২ নং]

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 04 & 05 No. Questions in are most important for the Examinee of 2025. ..... Editor***

Easy Math Basic Problem : Category-B & C

☑ Mean এর অংক

oo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–3; D.U. (Acc.) 2017 No–5; N. U. (Hon's) 3rd (Mgt) 2020 No–5**
The average marks of 39 students of a class are 50. The marks obtained by 40th student are 39 more than the average marks of all the 40 students. Find the mean marks of all the 40 students.

Solution : ব্যবস্থাপনা হাজেশন্স (২য় খণ্ড) বইয়ের ৬৭ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2016 No–05; 2022 No–7**
The average marks of 49 students of a class is 60. The marks obtained by 50th student is 49 more than the average marks of all the 50 students. Find the mean marks of all the 50 students.

Solution : ৬৭ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

ooo (99%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

oo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2013 No–4 (Marketing) 2018 No–6**
Marks obtained by 10 students are given below :
40, 30, 80, 70, 50, 20, 48, 95, 12, 88 compute the mean and median.

Solution : ৬৮ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2023 No–5**
The following set of data 13, 42, x, 7, 51, 69, 28, 33, 14, 8 has a median of 29; find out x.

Solution : ৬৮ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

ooo (99%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল-]

XX **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2024 No–5**
The arithmetic mean of two observations is 15 and their geometric mean is 9. Find out their harmonic mean.

Solution : ৫৬৩ পৃষ্ঠার **Question No-05** নং প্রশ্নের উত্তর দেখুন।

☑ Mode এর অংক

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2014 No–15; 3rd (Mgt) 2020 No–10**
Calculate the mode from the following data :

Income in Tk.	500–1000	1000–1500	1500–2000	2000–2500	2500–3000
No. of persons	13	11	10	8	5

Solution : ৭০ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

☑ Geometric Mean এর অংক

○○ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2011 No–11**

Calculate geometric mean of the data given below : —

Marks	0 – 10	10 – 20	20 – 30	30 ← 40	40 – 50
No. of students	3	4	6	3	2

Solution : ৭১ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

☑ Mean, Median, Mode, G.M, H.M and Nature of Distribution এর অংক

○○○ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2022 No– 10**

The following table gives the height (in inches) of 100 students of a class. Compute mean, mode and median of the height. Also comment about the nature of distribution :

Heig. (Inches)	60 – 62	62 – 64	64 – 66	66 – 68	68 – 70	70 – 72
No. of students	5	18	42	20	8	7

Solution : ৭৩ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2024 No–11**

Compute mean, median and mode from the following table. Comment about the nature of the distribution :

Height (inches)	60 – 62	62 – 64	64 – 66	66 – 68	68 – 70	70 – 72
No. of students	15	18	26	21	11	9

Solution : ৫৬৭ পৃষ্ঠার **Question No-11** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2018 No–11; (Mgt) 3rd 2019 No–10; 2021 No–17**

Find the following items from marks distribution :

Marks	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69	70 – 79	80 – 89	90 – 99
No. of Students	2	12	15	20	18	10	6	8

1. Mean, Median and Mode marks
2. Discuss the nature of distribution

Solution : ৭৪ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

☑ গড় গতিবেগের অংক

○○ **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2017 No–4**

A person travels 30 kms at 45 kmph, 20 kms at 30 kmph and 10 kms at 50 kmph. What is the average speed of that person?

Solution : ৭৭ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

☑ লুপ্ত গণসংখ্যার (Missing Frequency) এর অংক

○○ **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2017 No–12(b)**

The median and mode of the given frequency distribution are 34 and 32 respectively :

Class interval	20 – 25	25 – 30	30 – 35	35 – 40	40 – 45	45 – 50	Total
Frequency	15	15	?	?	?	17	120

Find the missing frequencies.

Solution : ৭৮ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2023 No–11**

Find out the missing frequencies from the incomplete table. When the median is 56. Total frequencies is 230.

Class interval	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90
Frequency	12	30	?	65	?	26	18

Solution : ৮০ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

০৫০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

ooo **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2021 No–11**

Find the missing frequencies from the incomplete table, when median is 45.

Class interval	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	Total
Frequency	5	15	?	30	25	?	5	120

Solution : ৮১ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

☑ Central/Middle Limit and Range এর % মান নির্ণয় :

ooo **Problem-15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2018 No–11**

From the following table showing the wage distribution in a factory :

Weekly wage (Tk.)	No. of employees
Less than 1,400	8
Less than 1,600	12
Less than 1,800	20
Less than 2,000	30
Less than 2,200	40
Less than 2,400	32
Less than 2,600	18
Less than 2,800	7
Less than 3,000	6
Less than 3,200	4

Determine :

(i) The wage limits for the middle 50% of the wage earners;
(ii) The percentage of the workers who earned between Tk. 1,750 and Tk. 2,250;
(iii) The percentage who earned more than Tk. 2,500 per week and
(iv) The percentage who earned less than Tk. 2,000 per week.

Solution : ৮৬ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2016 No–10**

The daily wages distribution of employees of a commercial institution is given below by the following table :-

Daily wages (Tk.)	No. of employees
250 – 300	25
300 – 350	30
350 – 400	55
450 – 500	67
500 – 550	39
550 – 600	14

(i) Find Mean and Median of the distribution
(ii) Find the central 50% wages limit of the employees
(iii) It was decided that the employee whose daily wages is Tk. 530 and more will be under tax deducted at source. What are the percentages of employees that are under tax scheme?

Solution : ৮৮ পৃষ্ঠার **Problem-17** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-17** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2017 No–10**

Calculate the mean and median from the following data :

Mid value	15	25	35	45	55	65	75	85
Frequency	5	9	13	21	20	15	8	3

Solution : ৯২ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

☑ Wrong Mean হতে Correct Mean এর অংক

○○ **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–8; (Accounting) 2017 No–7**

The mean of 200 was 50. Later on it was found that two observations were wrongly noted an 92 and 8 instead of 192 and 88. Find out the correct mean.

Solution : ৯৩ পৃষ্ঠার **Problem-20** নং প্রশ্নের উত্তর দেখুন।

☑ Combined Mean এর অংক

○○ **Problem-19** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2018 No–4**

The mean marks of 100 girls is 55, the combined mean is 58. What is the mean marks of 150 boys?

Solution : ৯৪ পৃষ্ঠার **Problem-23** নং প্রশ্নের উত্তর দেখুন।

☑ Less than, More than এবং মুক্ত প্রান্ত তথ্যের অংক

○○ **Problem-20** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2019 No–13**

The following data relates to the sales of 80 Companies :

Sales (Tk. lakhs)	No. of companies
Below 50	8
" 60	20
" 70	40
" 80	65
Above 90	80

Calculate the mean, median and modals sales.

Solution : ৯৬ পৃষ্ঠার **Problem-24** নং প্রশ্নের উত্তর দেখুন।

☑ প্রমাণসমূহ

○○ **Problem-21** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–4**

Proof that A.M ≥ G. M ≥ H.M.

Solution : ৯৭ পৃষ্ঠার **Problem-25** নং প্রশ্নের উত্তর দেখুন।

Chapter 03 Measures of Dispersion

(বিস্তার পরিমাপ)

Part-A : Brief Questions & Solutions :

☑ Measures of Dispersion

○○○ 01. **Define dispersion.** *[N.U. (Old) 2011, (Acc)- 2nd 2017]*

Ans : The distance of different values from the central value is called dispersion.

○○ 02. **How many types of measure of dispersion?**

Ans : Two types of measure of dispersion (i) Absolute measure (ii) Relative measure.

✪✪✪ 03. **Define interquartile range.** 99%

[N.U. (Acc)-2016, 2018]

Ans : The difference between the first and third quartile is called interquartile range.

✪✪✪ 04. **What is absolute measure?**

Ans : When dispersion is measured in original units then it is known as absolute measure of dispersion.

✪✪✪ 05 **What is mean deviation?** 99%

[Test Exam-Shaikh Burhanuddin College; N.U. (2nd)-2018 (Marketing)]

Ans : Mean deviation is the arithmetic mean of the deviation of series computed from any measures of central tendency.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪ 06. **What is quartile deviation.**

Ans : Quartile deviation is obtained by dividing the difference upper and lower quartile by 2.

✪✪ 07. **What is co-efficient of range?**

Ans : Co-efficient of range is the ratio of range to the sum of their highest value and the lowest value.

✪✪✪ 08. **Write the formula of co-efficient of variance.**

Ans : $CV = \frac{\text{Standard deviation (SD)}}{\text{Arithmetic Mean } \bar{X}} \times 100.$

✪✪✪ 09. **Deduce the formula of inter-quartile range.** 99% *[N.U. 2023 (Mgt)]*

Ans : The formula of inter-quartile range is $Q.D = Q_3 - Q_1$.

10. What is standard deviation? [২০২৪ সালে যা এসেছিল] [N.U. (Old) 2011, 2015; 2020, 2024 (Mgt)]

Or, Define standard deviation.

Ans : *The standard deviation is the positive square root of the mean of the squared deviations from their mean of a set of value.*

✪✪✪ 11. **Mention the relationship among mean, median and mode in case of positively skewed distribution.** *[N.U. (Old) 2012]*

Ans : In case of positive skewed distribution the relationship among mean, median and mode are $\bar{x} > me > mo$

✪✪✪ 12. **What is Box plot.** *[N.U. (Old) 2013]*

Ans : Box plot is the graphical presentation of five number summary.

✪✪✪ 13 **What are five number summary?** 99%

[N.U. BBA (Hon's) (3rd)-2017 (Mgt.)]

Ans : 1. Lowest value; 2. Highest value; 3. First Quartile; 4. Third Quartile; 5. Median.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 14. **What is five numbers summary?**

[N.U. (Old) 2012]

Ans : Five number summary especially construct with 5 value/ numbers. These are: (1) Smallest or, Minimum value-X_{min} (2) First quartile-Q_1 (3) Third Quartile-Q_3 (4) Median-Med (5) Largest or Maximum value-X_{max}

✪✪ 15. **What is range?**

[N.U. (Old) 2011; D.U. (Finance-2nd)2017]

Ans : Difference between highest value and lowest value.

✪✪✪ 16. **What is the co-efficient of determination?**

[N.U. 2018 (Acc)]

Ans : The numerical measure of degree of association between two variable x any y is called the co-efficient of correlation between them.

✪✪✪ 17. **What is price relative?** *[N.U. (Acc)-2016]*

Ans : A relative price is the price of a commodity scuh a goods or service in term of another the ratio of two prices.

✪✪ 18. **What is covariance?**

Ans : Covariance is a statistical measure that indicates the direction of the relationship between two variables.

✪✪ 19. **What is co-efficient of variation?**

Ans : Co-efficient of variance can be defined as the product of the ratio of standard deviation and mean. Symbolically,

$$CV = \frac{\sigma}{\bar{x}} \times 100$$

✪✪ 20. **What is variance?** *[N.U. (3rd)-2017(Mgt.)]*

Ans : Variance is one of the absolute measures of variation/dispersion. It is difined as the mean of sum of square of the difference between each value and its arithmetic mean.

✪✪✪21. **Write the formula of variance.** 99%

[N.U. (Acc)-2015; 2022 (Mgt.)]

Ans : Variance $= \frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2$

or, $\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2$

✪✪✪ 22 **Which measure is suitable for measures of dispersion?** 99% *[N.U. 2021 (Mgt.-3rd)]*

Ans : Standard deviation is suitable for measures of dispersion.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

☑ Skewness and Kurtosis

23. *What is of skewness? [২০২৪ সালে যা এসেছিল]*
[N.U. (2nd)- (Finance)-2018; (3rd)-2017, 2024 (Mgt.)]
Ans : Skewness is the degree of asymmetry or departure from symmetry in a distribution.

✪✪✪ 24. **What is moment?** *[N.U. (2nd)-2018 (Marketing)]*
Or, What is moment about origin?
Ans : The average of different powers of the observations is called a moment about origin.

✪✪✪ 25. **How many types of skewness?** 99%
[N.U. 2020 (Mgt.)]
Ans : Skewness is two types such as :
(i) Positive skewness, (ii) Negative skewness.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪ 26. **How many types of kurtosis?**
Ans : Kurtosis is three types such as :
(i) Mesokurtic, (ii) Leptokurtic, (iii) Platy kurtic

✪✪✪ 27. **What do you mean by positive skewness?**
[D.U. (2nd)-2017(Acc)]
Ans : Positve skewness means that the tail on the right side of a distribution is longer or more stretched out than the left side.

✪✪✪ 28. **Give the concept of Moments.**
Ans : Moments can be defined as the arithmetic mean of various powers of deviations taken from the mean of a distribution. Actually it describes the various characteristics of frequency distribution.

✪✪✪ 29. **What is kurtosis?** 99%
[N.U. BBA (Hon's) (Acc)-2017; 2021 (Mgt. 3rd)]
Ans : The word 'kurtosis' comes from a Greek word meaning 'humped'. In statistics, it refers to the degree of flatness or peakedness in the region about the mode of a frequency curve.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪ 30. **What is symmetric curve?**
Ans : Symmetric curve is the curve of a symmetric distribution which is bell shaped. It slopes equally from the top point to the both sides of the curve.

✪✪✪ 31. **What is the value of karl pearson's coefficient of skewness?**
Ans : Theoretically, the value of SK_p varies between ± 3. But for a moderately skewed distribution, value of $SK_p = \pm 1$.

✪✪ 32. **What is meso-kurtic?**
Ans : The frequency curve of a normal distribution is neither very peaked nor flat-topped i.e.; it is of moderate peakedness and known as mesokurtic.

✪✪✪ 33. **What does $\beta_2 > 3$ indicate?** *[N.U. (Acc)-2016, 2018]*
Ans : When $\beta_2 > 3$, the curve or distribution is leptokurtic.

✪✪ 34. **Which index number has no upward and downward tendency?** *[N.U. 2018]*
Ans : Cost of living index number has no upward and downward tendency.

✪✪✪ 35. **What are the minimum and maximum values of probability?** *[N.U. 2019]*
Ans : Minimum value 0 and maximum value 1.

Part-B & C : Short & Broad Questions :

☑ Measures of Dispersion

✪✪ 01. Define measures of dispersion. [উ : ব্যবস্থাপনা ইন্টারফেস (২য় খণ্ড) বইয়ের ১০৩ পৃষ্ঠার ৩.১১ নং]
Or, What do you mean by measures of dispersion? *[N.U. (Old) 2004, 2006, 2013]*

✪✪ 02. Discuss the advantages of standard deviation. [উ : ১০৪ পৃষ্ঠার ৩.১৭ নং] *[N.U. (Old) 2006, 2008, 2010]*

✪✪ 03. Write the mathematical properties of standard deviation. [উ : ১০৬ পৃষ্ঠার ৩.২৫ নং]
[N.U. (All) 2007, 2009; D.U. (3rd)-2016 (Mgt.)]

✪✪ 04. Mention the significance of measuring dispersion. [উ : ১০৩ পৃষ্ঠার ৩.১২ নং] *[N.U. (Old) 2007, 2009, 2014]*

✪✪ 05. Why standard deviation is considered to be the best measure? [উ : ১০৫ পৃষ্ঠার ৩.২০ নং]
[N.U. (All) 2014; N.U. (3rd)-2018 (Mgt.)]

✪✪ 06. Define dispersion. Discuss the different measures of dispersion. [উ : ১০১+১০৪ পৃষ্ঠার ৩.০১+৩.১৪ নং]
[N.U. (3rd)-2017, 2020 (Mgt.)]

✪✪✪ 07. **Distinguish between absolute and relative measures of dispersion.** 99% [উ : ১০৫ পৃষ্ঠার ৩.২১ নং]
[N.U. (Old) 2008, 2010, 2013; (Acc 2nd)-2016, 2018; (Hon's 3rd)-2021 (Mgt.)]
Or, Write down the differences between for absolute and relative measures of dispersion.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

08 **Write the properties of a good measure of dispersion.** 99% [Ans : Page No. 103; Ques. No 3.13]
[N.U. (Old) 2011, (Acc 2nd)-2015; (3rd)-2017, 2020, 2022); D.U. 2017 (Acc-2nd) 2017 (Managment-3rd]
Or, Write the qualilty of a good measure of disprsion.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০০ 09. **Show the differences between mean deviation and standard deviation.** 99%
[Ans : Page No. 105; Ques. No 3.22] *[N.U. (Hon's 3rd) 2022 (Mgt.)]*

☑ Skewness and Kurtosis

০০ 10. Define skewness? Discuss the types skewness? [উ : ১০৮ পৃষ্ঠার ৩.৩৫+৩.৩৬ নং]

০০০ 11. **Show the difference between skewness and kurtosis.** 99% [উ : ১১০ পৃষ্ঠার ৩.৪৩ নং]
Or, Distinguish between skewness and kurtosis. *[N.U. (Old) 2011, 2014; 2016 (Mgt.)]*

০০ 12. Distinguish between measures of dispersion and skewness. [উ : ১০৯ পৃষ্ঠার ৩.৩৯ নং]

013 **What is kurtosis? Describe the various types of kurtosis with diagram.** 99% [উ : ১১০ পৃষ্ঠার ৩.৪০ + ৩.৪১ নং] *[N.U. Mgt- 2019]*
Or, Define kurtosis? Discuss different types of kurtosis.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 07, 08 & 13 No. Questions in are most important for the Examinee of 2025.*** ***..... Editor***

Easy Math Basic Problem : Category-B & C

☑ গড়, মধ্যমা পরিমিত ব্যবধান, ভেদাংক ও বিভেদাংকের অংক

০০ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2018 No–7; D.U. (Accounting) 2017 No–6**

From the following data calculate standard deviation :

Wages (Tk.)	100	105	110	115	120	125	130	135
No. of Workers.	5	7	8	10	15	9	6	3

Solution : ব্যবস্থাপনা সাজেশন্স (২য় খণ্ড) বইয়ের ১১২ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2016 No–16(b)**

Find the standard deviation of following distribution :

Age	20 – 25	25 – 30	30 – 35	35 – 40	40 – 45	45 – 50
No.of persons	170	110	80	45	40	35

Solution : ১১৪ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2016 No–04; 2022 No– 09**

Give the following information's find standard deviation :
$N = 10, \sum x = 60, \sum x^2 = 1000.$

Solution : ১১৫ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2024 No–6**

From the following data calculate the standard deviation :
8, 10, 11, 9, 10, 11, 15, 10

Solution : ৫৬৩ পৃষ্ঠার **Question No-06** নং প্রশ্নের উত্তর দেখুন।

☑ চতুর্থক ব্যবধান ও চতুর্থক ব্যবধানাঙ্কের অংক

○○ **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–12(a)**

Calculate coefficient of variation and inter-quartile range from the following data :

Profit (Tk.)	10 – 19	20 – 29	30 – 39	40 – 49	50 – 59
No. of samples	8	12	20	6	4

Solution : ১১৭ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

☑ গড় ব্যবধান ও গড় ব্যবধানাংকের অংক

○○○ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2016 No–09; 2020 No–8**

From the following data calculate mean deviation :

16, 20, 22, 19, 18, 20, 30, 18, 20, 22.

Solution : ১২০ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

☑ একাধিক তথ্যের মধ্যে তুলনার অংক

○○○ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2023 No–12**

Compare the efficiency of two players from the given score :

Cricketer-A	98	17	0	29	70	45	3
Cricketer-B	26	32	53	35	26	17	52

Solution : ১২৩ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

○○○ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

○○ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2019 No– 11**

Prices of a particular commodity in five years in two cities are given below :

Price in city-A	Price in city-B
20	10
22	20
19	18
23	12
16	15

From the above data find the city which had more stable prices and why?

Solution : ১২৫ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2018 No–11**

Lives of two models of computer in a recent survey are given below :

Life in years	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25
IBM	5	13	20	7	5
DELL	2	7	12	20	9

Required :

(a) Which model has the more average life?

(b) Which model has the greater variation in life?

Solution : ১২৯ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2019 No–11**

The average weekly wages and standard deviations of two factories are as follows :

Factory	Average weekly wages (Tk.)	Standard deviation of wages	No. of workers
A	400	30	80
B	350	40	70

(i) Which factory pays larger amount for weekly wages?

(ii) Which factory shows larger variability in the distribution of wages?

Solution : ১৩১ পৃষ্ঠার **Problem-15** নং প্রশ্নের উত্তর দেখুন।

☑ সম্মিলিত তথ্যের অংক ও একাধিক তথ্যের মধ্যে তুলনার অংক

০০ **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–14 (Accounting) 2016 No–11; (Mgt) 3rd 2018 No–12**

The average weekly wages and standard deviation of two factories are as follows :-

Factory	Average weekly wages (Tk.)	Standard deviation	No. of workers
A	460	50	100
B	490	40	80

(i) Which factory pays larger amount of weekly wages?

(ii) Which factory shows larger variability in the distribution of wages?

(iii) What are the mean and standard deviation of all the workers in two factories taken together?

Solution : ১৩৩ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2021 No–9**

Two groups containing 200 and 300 observations. Their arithmetic mean is 15 and 20, standard deviation is 3 and 5 respectively. Calculate combined mean and standard deviation.

Solution : ১৩৭ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2014 No–12; 3rd (Mgt) 2016 No–12; 2021 No–12**

Particulars regarding income of two villages are given below :

	Village-A	**Village-B**
No. of people	600	500
Average income	175	186
Variance of income	100	81

Required :

(a) In which village is the variation in income greater?

(b) What is the total income of both the villagers put together?

(c) What is the average income of the people of both the villagers put together?

(d) What is the combined standard deviation?

Solution : ১৩৭ পৃষ্ঠার **Problem-20** নং প্রশ্নের উত্তর দেখুন।

☑ ভুল থেকে সঠিক গড়, পরিমিত ব্যবধান, ভেদাংক ও বিভেদাংকের অংক

০০ **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–12(b)**

The mean and the standard deviation of a sample of size 10. I were found to be 9.5 and 2.5 respectively. Later on, an additional observation become available. This was 15.0 and was included in the original sample. Find the mean and the standard deviation of 11 observation.

Solution : ১৪০ পৃষ্ঠার **Problem-23** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊠⊠ **Problem-15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2024 No–12**

The arithmetic mean and standard deviation of 100 items are found to be 30 and 10. But while calculating them two items are wrongly taken as 31 and 71 instead of 13 and 17. Find the correct mean and standard deviation.

Solution : ৫৬৮ পৃষ্ঠার **Question No-12** নং প্রশ্নের উত্তর দেখুন।

☑ পরিসর ও পরিসরাঙ্কের অংক

○○ **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2015 No–5**

Find the range and Co-efficient of range form the following data :-

Marks	20	30	40	50	60	70	80	90
No of Students	13	17	25	30	16	10	09	10

Solution : ১৪৩ পৃষ্ঠার **Problem-25** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-17** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2016 No–8**

The following are the wages (in Tk.) of 8 workers of a factory. Find the range of variation & also compute the co-efficient of range :

1400	1450	1520	1380	1485	1495	1575	1440

Solution : ১৪৩ পৃষ্ঠার **Problem-26** নং প্রশ্নের উত্তর দেখুন।

☑ Missing তথ্যের অংক

○○○ **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2018 No–10**

Find the missing information from the following:

	Group I	Group II	Group III	Combined
Number	50	?	90	200
Standard Deviation	6	7	?	7.746
Mean	113	?	115	116

Solution : ১৪৪ পৃষ্ঠার **Problem-27** নং প্রশ্নের উত্তর দেখুন।

☑ $\sum x$ ও $\sum x^2$ এর মান বের করার অংক

○○´ **Problem-19** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2019 No–8**

Mean of 200 items is 80 and their standard deviation is 10. Find the sum of the items and also the sum of squares of all the items.

Solution : ১৪৬ পৃষ্ঠার **Problem-29** নং প্রশ্নের উত্তর দেখুন।।

☑ বংকিমতার (Skewness) অংক

○○○ **Problem-20** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2016 No–6; 3rd (Mgt) 2022 No–8**

From the following data calculae Karl Pearson's skewness and make a comment :

Mean	0.2314
Median	0.25
Standard deviation	1.56

Solution : ১৪৯ পৃষ্ঠার **Problem-30** নং প্রশ্নের উত্তর দেখুন।

○○○ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

ooo **Problem-21** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2013 No–7; (Mgt) 3rd 2019 No–7;**

In a moderately skewed frequency distribution, the mean is Tk. 15 and the median is Tk. 14. If the co-efficient of variation is 30%, find the Pearsonian co-efficient of skewness.

Solution : ১৪৯ পৃষ্ঠার **Problem-32** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-22** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2016 No– 08; 2018 No–04**

For a group of 20 items $\sum x = 1452$ and $\sum x^2 = 144280$ and mode 63.7. Find the Pearsons coefficient of skewness.

Solution : ১৫১ পৃষ্ঠার **Problem-34** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-]

XX **Problem-23** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2024 No–9**

Find the coefficient of skewness from the given data :

Size	5	6	7	8	9	10
Frequency	14	16	17	21	17	15

Solution : ৫৬৫ পৃষ্ঠার **Question No-09** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-24** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2020 No–11**

Find out skewness of the following data and comment on the results :

Marks obtained	No. of students
40—50	8
50—60	12
60—70	20
70—80	25
80—90	15

Solution : ১৫৩ পৃষ্ঠার **Problem-36** নং প্রশ্নের উত্তর দেখুন।

ooo (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

oo **Problem-25** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2018 No–13**

Find out skewness of the following data and comment on the results :

Marks obtained	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90
No. of students	3	5	7	9	11	13	15

Solution : ১৫৬ পৃষ্ঠার **Problem-38** নং প্রশ্নের উত্তর দেখুন।

☑ সূচালতার (Kurtosis) অংক

oo **Problem-26** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2018 No–9**

The first four moments of a distribution are 0, 7.1, 26.2 and 408.7. Calculate kurtosis.

Solution : ১৬০ পৃষ্ঠার **Problem-41** নং প্রশ্নের উত্তর দেখুন।

☑ Moment, Skewness & Kurtosis এর অংক

oo **Problem-27** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2017 No–6; 2nd (Acc.) 2017 No–6; 2020 No–6**

If first four moments are 0, 7.1, 36.2 and 408.7 respectively. Calculate skewness and kurtosis and comments on the results

Or, From the following central moments calculate skewness and kurtosis and comment on the result :

$\mu_1 = 0, \mu_2 = 7.1\ \mu_3 = 36.2, \mu_4 = 408.7$

Solution : ১৬১ পৃষ্ঠার **Problem-43** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-]

⊠⊠ **Problem-28** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Acc.) 2016 No–12; 2020 No–14; 3rd (Mgt) 2020 No–17; 2022 No– 17; 2024 No–13**

Calculate skewness and kurtosis from the following information and comment on the results :

Population in family	2	3	4	5	6	7
Number of family	250	310	400	200	120	70

Solution : ১৬২ পৃষ্ঠার **Problem-44** নং প্রশ্নের উত্তর দেখুন।

✪✪ **Problem-29** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2019 No–7**

From the following central moments calculate skewness and kurtosis : M1 = 0, M2 = 9.5, M3 = 30.2, M4 = 350.8

Solution : ১৬৪ পৃষ্ঠার **Problem-46** নং প্রশ্নের উত্তর দেখুন।

Chapter 04 Probability and Probability Distributions
(সম্ভাবনা এবং সম্ভাবনা বিন্যাস)

Part-A : Brief Questions & Solutions :

✪✪✪ 01 **What is probability? 99%**

[Test Exam-Dhaka, Eden Girls (Acc.), New Model, AnandaMohan, B.L College; N.U. (All) 2011; (3rd)-2017(Mgt.)]

Ans : Probability is the favourable specific ratio of occurring or not occurring of any event.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 02. **What do you mean by mutually exclusive events?** *[N.U. 2011, (Acc)2016; (3rd)-2018 (Mgt.)]*

Ans : Events that can not happen together is called mutually exclusive event.

✪✪✪ 03. **What do you mean by r = 0?** *[D.U. (2nd) 2017(Acc)]*

Ans : The value of r = 0 means zero (0) correlation.

✪✪✪ 04 **What is meant by sample space.**

[Test Exam-Dhaka City, B.L College; N.U. (Old) 2011]

Ans : The set of all possible outcomes of an experiment is called sample space.

✪✪ 05. **What is binomial distribution?**

Ans : The distribution which contains only two possible outcomes (such as success or failure) is called binomial distribution.

✪✪✪ 06 **What is normal distribution? 99%**

[N.U. (3rd)-2022 (Mgt.)]

Ans : A distribution of a continuous random variable that has a single-peaked, bell-shaped curve is called a normal distribution.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 07. **Define joint probability. 99%**

[N.U. (2nd) 2013, 2017; D.U. 2017(Acc); (3rd)-2016 (Mgt.)]

Ans : The probability of two events occurring together or in succession is called joint probability.

✪✪✪ 08 **What is random variable? 99%**

[N.U. (Old) 2014, (Acc) 2017, N.U. (Finance)-2018; N.U. (3rd)-2018 (Mgt.)]

Ans : A random variable is a variable which assumes different values as a result of a random expriment.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪ 09. **What is conditional probability?**

[N.U. 2015, 2018(Acc)]

Ans : If two events A and B are dependent then the probability is known as conditional probability.

✪✪ 10. **What are equally likely event?**

Ans : The outcomes of a trial or events are said to be equally likely if each of them has an equal chance an of occurring, that is no outcome is more likely to occure than the others.

✪✪✪ 11. **Who invented mathematical measure of probability at first?**

Ans : Italian mathematician Galileo invented the mathematical measure of probability at first.

✪✪ 12. **What is experiment?**

Ans : The activity that results in, or produces and event is regarded as experiment.

✪✪✪ 13. **What is meant by event?** 99%
Or, What is event? *[N.U. Acc (3rd)-2019; 2022]*
Ans : One or more of the possible outcomes of doing something, or one of the possible outcomes from conducting an experiment.

✪✪✪ 14. **What is Venn Diagram?**
Ans : A pictorial representation of probability concepts in which the sample space is represented as a rectangle and the events in the sample space as portions of that rectangle.

✪✪ 15. **What is Probability Tree?**
Ans : Probability tree is a graphical representation showing the possible outcomes of a series of experiments and their respective probabilities.

✪✪✪ 16 **Illustrate the extreme values of probability.** 99% *[N.U. 2023 (Mgt.)]*
Ans : The extreme values of probability is 1.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪ 17. **What is discrete random variable?**
Ans : A variable that is allowed to take on only integer values.

✪✪✪ 18. **What is continuous random variable?**
Ans : A random variable which takes on an uncountable number of values within a range is called a continuous random variable.

✪✪ 19. **Find the range between 1 and –1.**
[N.U. BBA (Hon's) 2019(Acc) 3rd; 2021 (Mgt. 3rd)]
Or, What is the range between 1 and –1?
Ans : Range = Highest value – Lowest value.
= 1 – (– 1)
= 1 + 1
= 2

✪✪✪ 20. **Calculate range of the followings : 2, 3, 4, – 2, –4.** *[N.U. 2020 (Mgt.)]*
Ans. Range = Highest value – Lowest value
= 4 – (–2)
= 4 + 2
= 6

✪✪✪ 21. **Enunciate the parameters of Poisson distribution.** 99% *[N.U. 2023 (Mgt.)*
Ans : The parameters of Poisson distribution are m and λ.

✪✪ 22. **What is Poisson distribution?**
Ans : Poisson distribution is a discrete probability distribution and it is widely used in statistical work.

✪✪✪ 23 **Ascertain the range of – 3, 0, 3, – 5.** 99% *[N.U. 2023 (Mgt.)]*
Ans : Range = Highest value – Lowest value
= 3 – (–3) = 3 + 3 = 6.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

Part-B & C : Short & Broad Questions :

✪✪ 01. (a) What do you mean by probability?
[উ : ব্যবস্থাপনা হাইলাইটস (২য় খণ্ড) বইয়ের ১৬৬ পৃষ্ঠার ৪.০১ নং]
(b) Discuss the importance of probability in business decision making. [উ : ১৭১ পৃষ্ঠার ৪.২৫ নং]
[Test Exam :B.M, Edward, Chuadanga, Tularam college; N.U. (Old) 2007, 2010]

✪✪ 02. Describe the importance of normal distribution in statistical Analysis. [উ : ১৭২ পৃষ্ঠার ৪.২৮ নং]
[N.U. (Old) 2005, 2006, 2008, 2009]

✪✪✪ 03. **What are the properties of a binomial distribution?** 99% [উ : ১৬৯ পৃষ্ঠার ৪.১৮ নং]
[N.U. (Old) 2012; D.U. (3rd)-2016, 2017 (Mgt.)]

✪✪ 04. State the addition and multiplication theory of probability. [উ : ১৬৯+১৭০ পৃষ্ঠার ৪.১৯+৪.২০ নং]
[Test Exam : MM College; N.U. (Old) 2008, 2013, (Acc 2nd) 2015]
Or, Explain the addition and multiplication theorem of probability

✪✪✪ 05 **(a) Define binomial distribution.** 99% [উ : ১৬৭ পৃষ্ঠার ৪.০৬ নং] *[N.U. (Old) 2012; D.U. (3rd)-2016 (Mgt.)]*
(b) Show the distinction between continuous variable and discrete variable. 99%
[উ : ১৭০ পৃষ্ঠার ৪.২২ নং] *[N.U. BBA Hon's-2nd (Finance)-2018]*
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 06. State the main features of a normal distribution. [উ : ১৬৮ পৃষ্ঠার ৪.১৪ নং]
Or, Write down the properties of normal distribution. *[N.U. (Old) 2007, 2012; D.U. 2017 (Managment-3rd)*

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 03 & 05 No. Questions in are most important for the Examinee of 2025.*** ***..... Editor***

Easy Math Basic Problem : Category-B & C

☑ মুদ্রা (Coin) এর অংক

○○ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2018 No–2**

Find the probability of getting 2 heads in tossing 5 coins.

Solution : ব্যবস্থাপনা গাইডেন্স (২য় খণ্ড) বইয়ের ১৭৪ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2020 No–16**

Three unbiased coins are thrown at a time, construct a sample space. What is the probability of getting :
(i) At best two heads; (ii) At least two heads.

Solution : ১৭৫ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল]

☒☒ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2024 No–8**

A coin is tossed three times. What is the probability of getting a head in all the three tosses?

Solution : ৫৬৫ পৃষ্ঠার **Question No-08** নং প্রশ্নের উত্তর দেখুন।

☑ বল (Ball) এর অংক

○○○ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2023 No–8**

A bag contains 6 red and 8 green balls. If one ball is drawn at random, then what is the probability of the ball being green?

Solution : ১৭৬ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স (৯৯%) : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

○○○ **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2015 No–6; 3rd (Mgt) 2021 No–6**

There are 5 red and 4 white balls in a bag. One ball is drawn from the bag. What is the probability that it is either red or white?

Solution : ১৭৬ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2013 No–17; (Mgt) 3rd 2016 No–07**

A bag contains 7 red, 12 white and 4 green balls. What is the probability that.
(i) 3 balls drawn are all white and.
(ii) 3 balls drawn are one of each colour.

Solution : ১৭৮ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2015 No–14(b)**

There are 5 white and 7 red balls in a bag. Two balls are drawn such that a ball is drawn and replaced. What is the probability that a white ball and a red ball are drawn in that order? What would be the probability if the balls drawn were no put back into the bag?

Solution : ১৭৮ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন

○○ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2015 No–8**

A bag contains 5 blue, 4 red and 3 black balls. Two balls are drawn at random. Find the probability :-
(i) One black and blue
(ii) Not a red ball

Solution : ১৭৯ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2018 No–10**

A bag contains 10 red, 4 blue and 6 green balls. If 3 balls are drawn at random, determine the probability that :

(i) All 3 are red
(ii) All 3 are blue
(iii) At least 1 is blue
(iv) 2 are red and 1 is green
(v) 1 of each colour

Solution : ১৭৯ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

☑ তাস (Card) এর অংক

○○○ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2020 No–9**

Five cards are drawn from a packet of well-shuffled cards. Find the probability that 4 cards are aces.

Solution : ১৮৩ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

(৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

○○ **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–5; 3rd (Mgt) 2018 No–7**

4 cards are drawn from a pack of 52 cards without replacement. Find the probability of getting 3 aces.

Solution : ১৮৩ পৃষ্ঠার **Problem-17** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-12** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2022 No– 11**

3 cards are drawn at random from a pack of 52 cards. Find the probability of getting :

(i) Two kings
(ii) Two spades
(iii) One spade, One club and one diamond and
(iv) Two face cards.

Solution : ১৮৫ পৃষ্ঠার **Problem-20** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-13** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–14**

Four card are drawn at random from a pack of 52 cards. Find the probability that :

(a) They are a king, a queen, a jack and an ace.
(b) Two are kings and two are aces.
(c) All are diamonds.
(d) Two are red and two are black.
(e) There is one card of each suit.
(f) There are two cards of clubs and two cards of diamonds.

Solution : ১৮৫ পৃষ্ঠার **Problem-21** নং প্রশ্নের উত্তর দেখুন।

☑ দ্বিপদী বিন্যাস (Binomial Distribution) এর অংক

○○○ **Problem-14** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2017 No–8; 3rd (Mgt) 2018 No–6; 3rd (Marketing) 2018 No–4**

The mean of a binomial distribution is 40 and standard deviation is 6. Calculate the values of n, p and q.

Solution : ১৮৭ পৃষ্ঠার **Problem-22** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-15** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2017 No–13; 2019 No–14**

About 20% items of a production are defective in a certain factory. 10 products are packed in a packet. Among the 2,000 packets find the probability of getting packet containing—

(i) No defective product;

(ii) At least one product is defective;

(iii) At best one product is defective.

Solution : ১৮৯ পৃষ্ঠার **Problem-25** নং প্রশ্নের উত্তর দেখুন।

☑ পৈঁসো বিন্যাস (Poisson Distribution) এর অংক

○○○ **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2014 No–17; 2nd (Acc.) 2017 No–17; 2nd (Finance) 2019 No–17(b); (Mgt) 3rd 2018 No–15(b)**

A telephone operator receives 3 telephone calls on average per minute from 10 am to 12 am Find the probability that in a certain minute of the given time interval of a day, the operator receives:

(i) no call; (ii) at least two calls; (iii) at best two calls; (iv) two or three calls.

Solution : ১৯১ পৃষ্ঠার **Problem-27** নং প্রশ্নের উত্তর দেখুন।

○○○ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

☑ পরিমিত বিন্যাস (Normal Distribution) এর অংক

○○ **Problem-17** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2016 No–11**

The amount of money requested on loan applications at 'ABC Bank' follow the normal distribution, with a mean of Tk. 70,000 and standard deviation of Tk. 20,000. A loan application is received this morning. Find the probability that :

(i) The amount requested is Tk. 80,000 or more;

(ii) The amount requested is between Tk. 65,000 and Tk. 80,000.

Solution : ১৯৫ পৃষ্ঠার **Problem-28** নং প্রশ্নের উত্তর দেখুন।

☑ সম্ভাবনার বিবিধ অংক

○○ **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2016 No–8**

If A and b are two independent events and $P(A) = \frac{3}{5}$, $P(B) = \frac{2}{3}$, find $P(A \cap B)$ and $P(A \cup B)$.

Solution : ১৯৫ পৃষ্ঠার **Problem-29** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-19** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2017 No–15**

An industry has 10 male and 7 female workers. 5 workers selected at random. Find the probability of getting 3 female workers at least.

Solution : ১৯৬ পৃষ্ঠার **Problem-30** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-20** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2011 No–13; (Mgt) 3rd 2016 No–13; D.U. (Finance) 2017 No–15**

A manufacturing plant produces steel pipes in three plants with daily production volumes of 500, 1000 and 2000 units respectively. According to past experience, it is known that the fractions of defective output produced by the three plants are respectively at random 0.005,0.008 and 0.010. If a pipe is selected from a day's total production and found to be defective, find out (i) from which plant for this defective pipe, the probability is highest and (ii) what is the probability that it came from the first plant?

Solution : ১৯৭ পৃষ্ঠার **Problem-33** নং প্রশ্নের উত্তর দেখুন।

Chapter 05 Sampling & Sampling Distribution

(নমুনায়ন এবং নমুনায়ন বিন্যাস)

Part-A : Brief Questions & Solutions :

01. Describe the concept of sampling in brief. 99% *[N.U. (Acc)-2018; (3rd)-2021, 2023 (Mgt.)]*

Or, What is sampling?

Ans: The methods for selecting sample from a population as known as sampling.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

02. What is non sampling error?

Ans : Any sample estimate many also be affected by error is known as non sampling error.

03. What is population? 99%

[N.U. (All) 2011; Acc-2019; D.U. (3rd)-2016 (Mgt.)]

Ans : A population is the collection of all possible objects having certain characteristic under a study.

04. What is sample? *[N.U. (Finance)-2018]*

Ans : Sample is a portion of the elements of a population selected for direct measurement.

05. Define finite population.

Ans : A finite population is the aggregation of all the elements that are definite and countable under certain study.

06. Define infinite population.

Ans : If the aggregation of all the elements under a certain study is incompatible in number then the aggregation might be termed as infinite population.

07. What is parameters? *[N.U. (Old) 2011, (Acc)2015]*

Ans : The values that desirable the characteristics of a population are called parameters.

08. What is precision?

Ans : Precision is the degree of accuracy with which the sample mean can estimate the population mean.

09. What is cluster sampling? [২০২৪ সালে যা এসেছিল] *[N.U. 2022, 2024]*

***Ans :** Cluster sampling is a method of random sampling in which the population is divided into groups.*

10. What is sampling error? 99% *[Test Exam-Dhaka City, Eden (Acc.), Eden College(Mgt); N.U. 2nd (all) 2011; N.U. (3rd)-2018 (Mgt.); D.U. (2nd) 2017]*

Ans : The error that arises due to drawing inference about a population based on sampling is known sampling error.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

11. What is sampling fraction?

Ans : The fraction of population contained in a sample is called sampling fraction.

12. What is random sampling? 99% *[N.U. (Acc)2016]*

Ans : The random sampling methods are those in which each and every item in the population has an equal chance of being included in the sample.

13. Systematic sampling. *[D.U. (Finance)2017]*

Ans : The process of systematic sampling typically involves first selecting a fixed standing point in the larger population and then obtaning subsequent observation by using constant interval between sample taken.

14. What is degree of freedom?

Ans : The number of unrestricted chances for variation in the measurement being made.

15. What is small sample?

Ans : If the sample units in a population are less than 30, then the sample is said to be small sample.

16. What is large sample?

Ans : If the sample units in a population are more than 30 (n>30) the sample is said to be large sample.

17. What is sample survey? 99% *[N.U. (Acc) 2011]*

Ans : A technique of selecting a part of the population and collecting information from the selected individuals in known as sample survey.

18. What is sampling distribution?

Ans : A probability distribution consisting of all possible values of a sample statistic.

19. What is mean by sample point?

Ans : A single, individual outcome is called sample point.

Part-B : Short Questions :

✪✪ 01. Explain systematic random sampling. [উ : অনুশীলনী ইন্টারফেস (২য় খণ্ড) বইয়ের ২০৫ পৃষ্ঠার ৫.২৪ নং]

[Test Exam : Kobi Nzarul, Sathkhira College; N.U. (Old)-2011]

✪✪ 02. What are the causes of sampling error and non-sampling error? [উ : ২০২+২০৩ পৃষ্ঠার ৫.১৪ + ৫.১৭ নং]

[D.U. (3rd)-2016 (Mgt.)]

✪✪✪ 03 **Show the difference between population and sample.** 99% [উ : ২০১ পৃষ্ঠার ৫.০৮ নং]

[Test Exam : Azizul, Sadat, Ralngur, Satkhir college; N.U. (3rd)-2018 (Mgt.)]

Or, State the differences between population and sample.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Part-C : Broad Questions :

✪✪✪ 01. Explain the importance of sampling in solving business problem. [উ : ২০৫ পৃষ্ঠার ৫.২৩ নং]

[Test Exam : Khulna, Rajshahi, B.M., Munsur, Tongi college; N.U. (Old) 2004]

✪✪ 02. Distinguish between sampling and non sampling error. [উ : ২০৩ পৃষ্ঠার ৫.১৮ নং]

✪✪ 03. Describe the two important principles of sampling. [উ : ২০৫ পৃষ্ঠার ৫.২৫ নং]

[N.U. (Old)2004, 2009; D.U. 2017 (Finance 2nd)]

✪✪✪ 04. **Discuss about the various sampling method.** 99% [উ : ২০৪ পৃষ্ঠার ৫.১৯ নং]

[Test Exam : Comilla College; N.U. (All) 2006, 2010; (3rd)-2018 (Mgt.)]

Or, Discuss the various methods of Sampling.

✪✪✪ 05 **(a) Define population, sample and sampling with example.** 99%

[Ans : Page No. 200+201; Ques. No 5.01+5.05+5.07] *[N.U. Mgt- 2019, 2022]*

(b) Explain the advantages of sample survey compared to population survey. 99%

[Ans : Page No. 200; Ques. No 5.04] *[N.U. Mgt- 2019, 2022]*

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Student's, In above Suggestions- 03 No. in Part-B and 05 No. Questions in Part-C are most important for the Examinee of 2025.*** ***– Editor***

Easy Math Basic Problem : Category-B & C

✪✪ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–9**

Draw a sample of 3 units from a population of (2, 4, 6, 8). Write down samples. Show that variance mean of samples is equal to population variance.

Solution : অনুশীলনী ইন্টারফেস (২য় খণ্ড) বইয়ের ২০৬ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

✪✪ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–6**

The mean age of a combined group of men and women is 30 years. If mean age of the group of men is 32 and that of the group of women is 25, find out the percentage of men and women in the group.

Solution : ২০৭ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-]

✪✪ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. 2nd (Old) 2012 No–15; D.U. Affiliated (Acc.) 2017 No–16; N.U. (MBA Final) Mgt 2016 No– 07; (Hon's) 3rd (Mgt) 2016 No–15; 2024 No–15**

In a simple random sampling of 600 men taken from a village in which 400 men are found to be farmers. In another simple random sampling of 900 men taken from another village in which 450 men are found farmers. Do the data indicate that there is a significant difference in the occupation of men in the two villages?

Solution : ২০৮ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2013 No–13; (Acc.) 2017 No–13; 3rd (Marketing) 2018 No–12**

Dhaka City Motors/Progoty Motors/Navana Motors Limited has 5 representative. The numbers of sold cars in the last month are given :

Representative	A	B	C	D	E
Production	5	15	10	25	20

(i) How many different samples are possible to take considering 2 at a time?
(ii) Record all possible samples and find its mean taking 2 at a time.
(iii) Compare between mean of sample means and population mean?

Solution : ২১০ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

Chapter 06 Hypothesis Testing

(কল্পনা যাচাই)

Part-A : Brief Questions & Solutions :

০০০ **01. Define hypothesis?** 99%

Ans : Hypothesis is a statement about a population parameter developed for the purpose of testing.

০০ **02. What is statistical hypothesis?**

Ans : A statistical hypothesis is a statement about the probability distribution of a set of random variables.

03. What is null hypothesis? [২০২৪ সালে যা এসেছিল]

Or, Define null hypothesis.

[Test Exam-Dhaka City (Acc.), Eden College (Mgt); N.U. 2011; 2018, 2020, 2023, 2024 (Mgt.)]

Ans : The hypothesis about a population parameter that we wish to test is called the null hypothesis.

০০০ 04 **What is critical value?** 99%

Or, Define critical value. *[Test Exam-Eden College N.U. (3rd)-2018, 2020, 2022 (Mgt.)]*

Ans : The value of the standard statistic (Z or t) beyond which we reject the null hypothesis.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০০ **05. What is power of the test?** *[N.U. (Finance)-2018]*

Ans : The probability of not committing a type II error is called the power of the test hypothesis.

০০ **06. What is two-tailed test?** *[N.U. (Acc)-2015]*

Or, Define two tailed test.

Ans : When the hypothesis about the population mean is rejected for value of falling into either tail of the sampling distribution, it is called two tailed test.

০০০ 07 **Define type-1 error.** 99%

Or, What is meant by Type 1 Error?

Or, Type -1 Error. *[D.U. (Finance)-2017; (3rd)-2016 (Mgt.); N.U. Acc 3rd-2019; 2022]*

Ans : Type 1 error is the rejecting a null hypothesis when it is true.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০ **08. What is test of hypothesis?**

Ans : Test of hypothesis is a statistical procedure that is used to provide evidence in favour of same statement.

০০০ **09. What is Alpha?**

Ans : The probability of Type 1 error.

০০০ 10 **Define Type II Error.** 99%

[N.U. (All) 2013; N.U. (3rd)-2017 (Mgt.)]

Ans : Type II error is the accepting a null hypothesis when it is false.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

০০০ **11. What is chi-square (χ^2) test?** 99%

[N.U. (Acc) 2015; D.U. (2nd)-2017 (Acc); (3rd)-2016 (Mgt.)]

Ans : $\chi^2 = \sum \frac{(O-E)^2}{E}$

০০০ **12. What is significance level?** *[N.U. 2017, 2018(Acc)]*

Ans : The Probability of rejecting the null hypothesis given that it is true (type I error) and is most of set at 0.5 (5%)

13. What is T-test? [২০২৪ সালে যা এসেছিল]

[N.U. 2024 (Mgt. 3rd)]

Ans : T-test is a statistical hypothesis test used to determine whether there is a significant difference between the means of one or two groups when the sample size is small or less than 30 and the population standard deviation is unknown.

Part-B & C : Short & Broad Questions :

✪✪ 01. What is hypothesis testing? [উ : ব্যবস্থাপনা ইন্টারফেস (২য় খণ্ড) বইয়ের ২১২ পৃষ্ঠার ৬.০১ নং] *[N.U. (Old) 2006, 2007, 2009, 2010]*

✪✪✪ 02. **What is T-test.** 99% [উ : ২১৫ পৃষ্ঠার ৬.১৬ নং] *[N.U. (Old) 2007, 2010]*

✪✪ 03. Discuss the procedure of testing a hypothesis. [উ : ২১২ পৃষ্ঠার ৬.০৬ নং] *[N.U. (Old) 2011]*

✪✪✪ 04. **Write short note :** 99% [উ : ২১৩ পৃষ্ঠার ৬.০৭ নং] *[N.U. 2018(Acc-2nd)]*

(a) Hypothesis and Hypothesis testing;

(b) Two way classification of analysis of variance;

✪✪✪ 05. (a) What do you mean by test of hypothesis? [উ : ২১৩ পৃষ্ঠার ৬.০৯ নং] *[N.U. (Old) 2009]*

(b) Describe the various steps involved in testing hypothesis. [উ : ২১৯ পৃষ্ঠার ৬.৩৫ নং]

Or, What steps are the followed in hypothesis testing? *[N.U. BBA Hon's-2nd (Finance)-2015, 2018]*

✪✪✪ 06 **(a) What is chi-square test?** 99% [উ : ২১৬ পৃষ্ঠার ৬.১৮ নং]

(b) Discuss three conditions for the application of chi-square test. 99% [উ : ২১৬ পৃষ্ঠার ৬.১৯ নং]

[Test Exam : Satkhira, Sadat, Ananda Mohan, Thakurgang; N.U. (Old) 2005, 2006, 2007, 2008, 2010]

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 04 & 06 No. Questions in are most important for the Examinee of 2025.*** ***..... Editor***

Easy Math Basic Problem : Category-B & C

✪✪ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2017 No–6**

A random sample of 118 women gave a mean weight of 50 kg with a standard deviation of 8 kg. Test the hypothesis that the mean weight in the population is 40 kg.

Solution : ব্যবস্থাপনা ইন্টারফেস (২য় খণ্ড) বইয়ের ২২১ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

✪✪✪ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–09; N.U. 3rd (Mgt) 2016 No–7; 2021 No–8; 3rd (Marketing) 2018 No–13**

Intelligence test-on groups of boys and girls gives the following result :

	Mean	S.D	N
Girls	75	15	150
Boys	70	20	250

Is there a significant difference in the mean scores obtained by boys and girls?

Solution : ২২৩ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

✪✪ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 4th (Accounting) 2018 No–9**

The sample mean of two groups of 1300 and 1600 sizes are 69.59 and 69.75. The standard deviations are 2.6 and 3 respectively. Test the difference between two sample groups at 5% significant level.

Solution : ২২৫ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

✪✪✪ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2018 No–17**

A buyer of electric bulbs bought 100 bulbs each of two famous brands. 'Brand A' had a mean life of 1500 hours with a standard deviation of 50 hours, whereas 'Brand B' had a mean life of 1530 hours with a standard deviation of 60 hours. Can it be concluded at 5% level of significance that the two brands differ significantly in terms of their quality?

Solution : ২২৬ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's)4th (Accounting) 2020 No–17**

Two salesmen Mukit and Azad are working in a company. From the following data, state whether there is any significant difference in the average sales between two salesmen :

Salesmen	Mukit	Azad
No. of sales	30	27
Average sales (in Taka)	205	308
Standard deviation (in Taka)	30	38

Solution : ২২৮ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Marketing) 2018 No–14**

Mean and standard deviation of production of the two worker groups are given below :-

	1st group	2nd group
Number of workers	17	10
Mean of production	50 mound	55 mound
Standard deviation	6 mound	5 mound

Test the significant difference between the two groups.

Solution : ২২৯ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–15; 3rd (Mgt.) 2021 No–15; 4th (Acc.) 2019 No–14(b); (MBA Final) Mgt. 2016 No– 07; D.U. Affiliated (Acc.) 2017 No–16;**

In a random sample of 600 men taken from a big city in which 400 are found to be smokers. In another simple random, sample of 900 men taken from another city in which 450 are found smokers. Do the data indicate there is a significant difference in the habit of smoking in the two cities?

Solution : ২৩১ পৃষ্ঠার **Problem-15** নং প্রশ্নের উত্তর দেখুন।

○○ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2016 No–17**

A random sample 234 students from 450 students in Dhaka University and 171 students from 380 students in Rajshai University gave their opinion against smoking. Is the difference of opinion of two University students are significant at 5% significance level?

Solution : ২৩৩ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Acc.) 2015 No–16(b); 2017 No–17; 3rd (Marketing) 2017 No–15; 2022 No– 16**

Price of shares (In Tk.) of a company on the different days in a month were found to be :
66, 65, 69, 70, 69, 71, 70, 63, 64, 68
Test whether the mean price of the shares in the month is Tk. 65.

Solution : ২৩৪ পৃষ্ঠার **Problem-20** নং প্রশ্নের উত্তর দেখুন।

○○○ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Acc) 2016 No–9; 3rd (Mgt) 2023 No–16**

A simple random sample of size 36 is drawn from a finite population consisting of 101 units. The population's standard deviation is 12.60. Find the standard error of sample mean when the sample is drawn : (a) With replacement; (b) Without replacement.

Solution : ২৩৬ পৃষ্ঠার **Problem-23** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

Chapter 07 Correlation & Regression Analysis

(সহ-সম্বন্ধ ও নির্ভরণ বিশ্লেষণ)

Part-A : Brief Questions & Solutions :

☑ Correlation Analysis

01. What is correlation? [২০২৪ সালে যা এসেছিল] [N.U. 2024 (Mgt. 3rd)]

Ans : The term correlation mean the combination of two interrelated on dependable variance. Practically the interrelation between two or more variables are called correlation.

✪✪✪ 02. **What is coefficient of correlation?** 99%

[Test Exam-Eden (Acc.), Eden (Mangé), Dhaka Colege, New Model College; N.U. (All) 2011; (2nd)-2018 (Marketing); (3rd)-2018, 2021 (Mgt.); Acc 3rd-2019; D.U. 2017 (Finance)]

Or, Define co-efficient of correlation.

Or, Conflation coefficient.

Or, What is correlation?

Or, What is meant by correlation?

Ans : Co-efficient of correlation is the measure of the strength of the linear relationship between two variables.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪ 03. **In case of unequal rank, write the formula of rank correlation.**

Ans : Rank correlation $1 - \frac{6\Sigma d^2}{n(n^2-1)}$

✪✪✪ 04. **What is rank correlation?** 99%

[N.U. 2020 (Mgt.)]

Ans : The co-efficient of correlation between two sets of rank is called rank correlation.

✪✪ 05. **Why correlation is used?**

Ans : "Correlation analysis attempts to determine the degree of relationship among variables".

✪✪✪ 06. **Define zero correlation?**

Ans : When there is no correlation between two variables is called zero correlation.

☑ Regression Analysis

✪ 07. **What are the different types of correlation?**

Ans : Three types correlation.

(i) Positive correlation;

(ii) Negative correlation;

(iii) Zero correlation.

✪✪✪ 08. **What is regression line?**

Ans : A line fitted to a set of data points to estimate the relationship between two variables.

✪✪✪ 09. **What is regression model?** *[N.U. (Finance)-2018]*

Ans : A regression model is used to investigate the relationship between two or more variable and estimate one variable based on the others.

✪✪✪ 10. **What is regressions analysis?** 99%

[N.U. (Old) 2011; 2021 (Mgt. 3rd)]

Ans : Regression analysis is a mathematical measure of the average relationship between two or more variables in terms of original units of the data.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

11. What is independent variables? [২০২৪ সালে যা এসেছিল] [N.U. 2022, 2024 (Mgt. 3rd)]

Ans : The known variable in regression analysis is called independent variables.

✪✪ 12. **What do you mean by regression?** *[N.U. BBA (Hon's) (2nd)-2017 (Marketing;)D.U. (2nd)-2017 (Marketing)]*

Ans : Regression is the general process of predicting one variable, from another, using statistical method and previous data.

✪✪✪ 13. **Define regression coefficient.** *[N.U. (Acc) 2016]*

Ans : The algebraic measurement of the effect of independent variable upon dependent variable is called coefficient of regression.

✪✪✪ 14. **What is dependent variable?** *[N.U. (Old) 2013]*

Ans : The variable we are trying to predict in regression analysis is called dependent variable.

✪✪✪ 15. **What is least square method?** 99%

[N.U. (2nd)-2018 (Marketing)]

Ans : The special mathematical model that is used to construct trend is called least squares method.

✪✪✪ 16. **According to least square method write the formula of co-efficient of correlation.**

Ans : $r_{xy} = \sqrt{b_1 \times b_2}$ Here, b_1 and b_2 are regressing co-efficient.

✪✪✪ 17. **What is scatter diagram?** 99%

[Test Exam-Dhaka (Acc.), Dhaka Commerce, Habibullahbaher, Shaikh Burhanuddin College; N.U. (All) 2014, (Acc) 2017; (3rd)-2018, 2020 (Mgt.); D.U. (3rd)-2016 (Mgt.)]

Ans : The simplest device for studying correlation in two variables is a special type of at chart called scatter diagram.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪ **18. Give a pen-picture of scatter diagram.**
[N.U. 3rd (Mgt) 2013]

Ans : Let,

Capital (in lac Tk.) : X	5	10	15	20	25
Profit (in lac Tk.) : Y	3	4	8	12	18

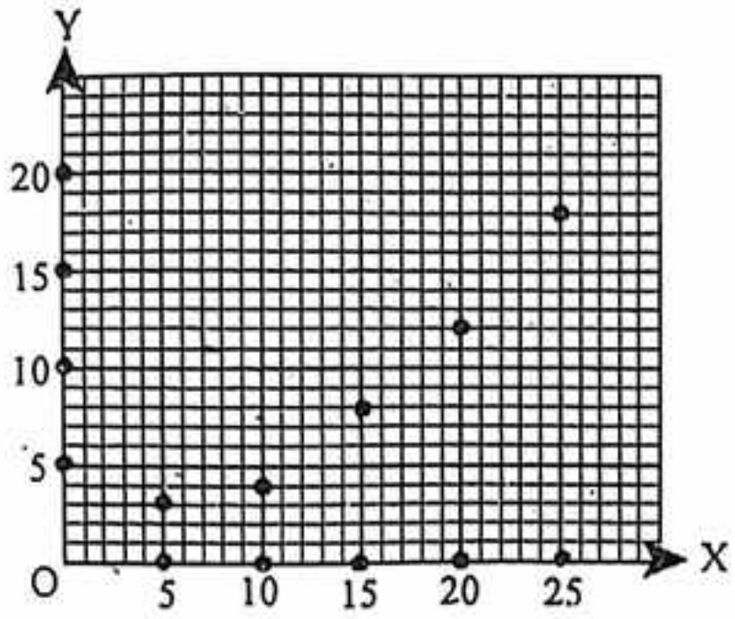


Figure : Scatter diagram

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 19. **What is empty set?** *[N.U.. (Finance)-2018]*
Ans : The empty set is unique set having no elements its size or cardinality is zero.

✪✪✪ 20. **What is meant by r = + 1?** *[N.U. 2021 (Mgt. 3rd)]*
Or, r = + 1, what does it mean?
Ans : r = 1 means perfect positive correlation between x and y.

✪✪ 21. **What is ment by r = 0.75?**
Ans : It is means by the partial positive correlation between the two variable. In this case, the change of the two variables is some direction.

✪✪✪ 22. **If $b_{yx} = -0.8$ and $b_{xy} = -0.2$, r_{xy} =?**
[Test Exam-Dania College; N.U. (Old) 2013]
Ans : $r_{xy} = \sqrt{b_{yx} \times b_{xy}}$
$= \sqrt{(-0.8) \times 0 (-0.2)}$
$= 0.4$

✪✪✪ 23. **What is meant by r = 1, r = – 1 and r = 0?** 99%
[N.U. 2nd (All) 2012, 2014, (Acc) 2017; (3rd)-2017 (Mgt.); Acc 3rd-2019]
Ans : The value of r = 1, means the perfect positive correlation.
The value of r = –1, means the perfect negative correlation.
The value of r = 0 means the zero (0) correlation.

✪✪ 24. **What is difference between T-test and Z test?**
Ans : Is the sample size is small (n < 30) and population standard deviation (σ) is unknown then t-test is used. On the other hand Z-test is used for the large sample (n < 30) test.

✪✪✪ 25. **Which test is known as small test?** 99%
[N.U. BBA (Hon's)2021 (Mgt. 3rd)]
Ans : T-test is known as small test. Another name of t-test is student test.
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 26. **What is meant by multiple regression?**
Ans : The statistical process by which more then two variables are used to predit another variable. *[N.U. (Acc) 2015]*

✪✪✪ 27. **What is slope?**
Ans : A constant for any given straight line whose value represents how much each unit change of the independent variable changes the dependent variable.

Part-B & C: Short & Broad Questions :

☑ Correlation Analysis

✪✪ 01. What is Correlation? [উ : ব্যবসায়িক ব্যবস্থাপনা ইন্সপেক্স (২য় খণ্ড) বইয়ের ২৩৮ পৃষ্ঠার ৭.০১ নং]
Or, Defination of correlation. *[Test Exam : Shahid Sirajuddin College; N.U. (Marketing)- 2nd 2015]*

✪✪✪ 02. What is rank correlation? [উ : ২৩৯ পৃষ্ঠার ৭.০৫ নং] *[N.U. 2018 (Acc-2nd)]*
Or, Defination of Rank Correlation

✪✪ 03. Write down the significance of measuring correlation. [উ : ২৪১ পৃষ্ঠার ৭.০৭ নং]
[Test Exam : Sadat, Shahid Munsur, Thakurgaw, Ananda Mohan College; N.U. (Old) 2011, 2014]
Or, Discuss the importance or uses or utilities of correlation analysis.

✪✪✪ 04. **What are the properties of co-efficient of correlation?** 99% [উ : ২৪১ পৃষ্ঠার ৭.০৯ নং]
[N.U. BBA (Hon's) BBA Hon's-2nd (Finance)-2018]

☑ Regression Analysis

✪✪✪ 05. What is regression? [উ : ২৪২ পৃষ্ঠার ৭.১০ নং] *[N.U. (Old) 2004, 2009, (Marketing-2nd) 2015]*
Or, What is the concept of regression?

✪ 06. State the use of regression analysis. [উ : ২৪২ পৃষ্ঠার ৭.১১ নং]

✪✪ 07. (a) What is regression co-efficient? [উ : ২৪৩ পৃষ্ঠার ৭.১৫ নং] *[N.U. (Old) 2005]*
(b) State some of the important properties of regression co-efficient. [উ : ২৪৫ পৃষ্ঠার ৭.২২ নং]
[Test Exam : Lalmatia, Rangpur, B.M. Sultan.college; N.U. (Old) 2005, 2015]

✪✪✪ 08. **State the relationship between co-efficient of correlation and regression co-efficient.** 99%
[উ : ২৪৫ পৃষ্ঠার ৭.২০ নং] *[D.U. (3rd)-2016 (Mgt.)]*

✪✪✪ (09) **Show the differences between correlation and regression.** 99% [উ : ২৪৫ পৃষ্ঠার ৭.২১ নং]
Or, Distinguish between correlation and regression. *[N.U. (3rd)-2017, 2019 (Mgt.)]*
[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 04, 08 & 09 No. Questions in are most important for the Examinee of 2025. ..... Editor***

Easy Math Basic Problem : Category-B & C

Correlation Analysis

☑ কার্ল পিয়ারসনের সূত্রের সাহায্যে সংশ্লেষাংকের অংক

✪✪ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–11 (Accounting) 2016 No–13**

From the following data find the coefficient of correlation between age the playing habit of students :

Age in years	15	16	17	18	19	20
No. of students	250	200	150	120	100	80
Regular players	200	150	90	48	30	12

What that conclusion do you draw from the result?

Solution : ব্যবস্থাপনা ইন্সপেস (২য় খণ্ড) বইয়ের ২৪৮ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

✪✪ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2019 No– 12**

Find the co-efficient of correlation (r) between X and Y from the following data:

X	3	6	5	4	4	6	7	5
Y	3	2	3	5	3	6	6	4

Also calculate the regression of Y on X and predict the average value of Y when X is 9.

Solution : ২৫০ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

✪✪ **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2016 No–6**

Calculate the coefficient of correlation between X and Y series from the following data :-

$N = 12;\ \Sigma X = 120;\ \Sigma Y = 130;\ \Sigma(X-8)^2 = 150;$

$\Sigma (Y-10)^2 = 200$ and $\Sigma (X-8)(Y-10) = 50$.

Solution : ২৫২ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

✪✪✪ **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2023 No–15**

A researcher collected the following information for two variables x and y;

No. of pairs, $n = 20$, $r = 0.5$, $\bar{x} = 15$, $\bar{y} = 20$, $\sigma_x = 4$, $\sigma_y = 5$. Later on, it was found that one pair of value has been wrongly taken as : $x = 16$, $y = 30$; where as the correct values were : $x = 26$, $y = 35$. Find the correct value of r.

Solution : ২৫৪ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

✪✪✪ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০ **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–13(b)**

From the following table, calculate the coefficient of correlation by Karl Pearson's method :

Marks obtained by Saniyat	6	2	10	4	8
Marks obtained by Deep	9	11	?	8	7

Arithmetic means of series are 6 and 8 respectively.

Solution : ২৫৬ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-06** ***Year Question & Solution*** ⊙ **D.U. BBA (Hon's) 2nd (Marketing) 2017 No–4; N.U. BBA (Hon's) 3rd (Mgt) 2020 No–13**

Find out the co-efficient of correlation of the following data and comment on the result:

X (Income)	50	720	202	130	140	182
Y (Expenditure)	32	378	105	86	84	72

Solution : ২৫৭ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

⊠⊠ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2024 No–7**

Calculate correlation between price and demand from the following data :

Price (Tk.)	3	5	6	8	9
Demand (Unit)	3	4	5	7	8

Solution : ৫৬৪ পৃষ্ঠার **Question No-07** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2017 No–7**

You are give the following 3 pairs of data (1, 3), (2, 1) and (1, 6). Determine co-efficient of correlation.

Solution : ২৫৮ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2014 No–13**

Following figures give the rainfall in inches for the year and the production in mounds for Rabi and Kharif crop. Calculate the Karl Pearson's co-efficient of correlation between rainfall and total production :

Rainfall	20	22	24	26	28	30	32
Rabi	16	18	20	32	40	38	35
Kharif	16	17	20	16	20	22	20

Solution : ২৬০ পৃষ্ঠার **Problem-11** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2015 No–12**

The data relating to capital and profit of 5 shops are given :–

Capital (in lac Tk.) : X	5	10	15	20	25
Profit (in lac Tk.) : Y	3	4	8	12	18

(i) Draw a scatter diagram.

(ii) Compute the co-efficient of correlation and interpret its value.

Solution : ২৬১ পৃষ্ঠার **Problem-12** নং প্রশ্নের উত্তর দেখুন।

০০০ Problem-11 *Year Question & Solution* ⊙ N.U. BBA (Hon's) 3rd (Mgt) 2023 No–13

The data relating to sales and profit of the five shops are given :

Sales (In lakh Tk.)	10	20	30	40	50
Profit (In lakh Tk.)	3	4	8	12	17

You are required to :

(i) Draw a scatter diagram;

(ii) Compute the co-efficient of correlation and interpret its value.

Solution : ২৬২ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০ Problem-12 *Year Question & Solution* ⊙ N.U. BBA (Hon's) 2nd (Accounting) 2019 No–14

Calculate the Karl Pearson's co-efficient of correlation and probable error between the value of X and Y :

x	78	98	96	69	59	79	68	61
y	125	137	128	112	107	115	110	108

Solution : ২৬৭ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

০০ Problem-13 *Year Question & Solution* ⊙ N.U. BBA (Hon's) 2nd (Marketing) 2017 No–17

The given data is:

X	7	6	5	4	3	2	1
y	18	16	14	12	10	6	8

(a) Calculate Karl Pearson's coefficient of correlation.

(b) Find the standard error of the coefficient of correlation.

(c) Determine the probable error 'r'.

Solution : ২৬৮ পৃষ্ঠার **Problem-19** নং প্রশ্নের উত্তর দেখুন।

☑ Ranking পদ্ধতিতে সংশ্লেষাংক নির্ণয়

০০০ Problem-14 *Year Question & Solution* ⊙ N.U. BBA (Hon's) 3rd (Mgt) 2017 No–12

Two teachers examined 10 answer scripts. They gave the marks 10 as follows :

Script No.	1	2	3	4	5	6	7	8	9	10
1st Examiner	70	50	75	74	70	30	25	50	60	40
2nd Examiner	65	52	75	70	75	20	33	45	62	30

Compute the rank co-efficient of correlation marks given by the both examiners and comment.

Solution : ২৭২ পৃষ্ঠার **Problem-21** নং প্রশ্নের উত্তর দেখুন।

০০০ Problem-15 *Year Question & Solution* ⊙ N.U. BBA (Hon's) 3rd (Mgt) 2022 No– 06

From the following data calculate the Rank Co-efficient of correlation :

x	4	6	5	8	10
y	2	1	3	5	8

Solution : ২৭৪ পৃষ্ঠার **Problem-23** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০০ **Problem-16** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2017 No–14**

The data of aptitude score and productivity are given below :—

Aptitude score	9	18	19	20	21	23	22	24	26	25
Productivity index	33	23	30	42	29	32	39	46	40	44

(a) Find rank correlation co-effient between aptitude score and productivity index and comment.

(b) Interpret the result.

Solution : ২৭৫ পৃষ্ঠার **Problem-24** নং প্রশ্নের উত্তর দেখুন।

Regression Analysis

☑ x এর উপর y এর নির্ভরণ সমীকরণের অংক

০০ **Problem-17** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2011 No–15 (Accounting) 2017 No–16**

Compute the appropriate regression equation for the following data :-

x (Independent Variable)	y (Dependent Variable)
2	18
4	12
5	10
6	8
8	7
11	5

Solution : ২৭৭ পৃষ্ঠার **Problem-25** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-18** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2013 No–14; D.U. 3rd (Accounting) 2017 No–14; N.U. 3rd (Mgt) 2021 No–13**

Capital and Profit of a firm are shown in the following table :

Capital (cror Tk.)	5	10	15	20	25
Profit (lakh Tk.)	3	4	8	12	18

(i) Determine the co-efficient of correlation between profit and capital.

(ii) Regression equation of profit on capital.

(iii) What should be the profit when capital is Tk. 32 lac.?

Solution : ২৭৮ পৃষ্ঠার **Problem-26** নং প্রশ্নের উত্তর দেখুন।

☑ এক সাথে দুটি নির্ভরণ সমীকরণের অংক

০০ **Problem-19** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2015 No–13**

A researcher wants to find out if there is any relationship between the ages of the husbands and the ages of the wives. He took a random sample of 7 couples whose respective ages are given below :–

Age of husband (x)	25	27	29	32	35	37	39
Age of wife (y)	18	20	20	25	25	30	37

Compute the two regression equations and determine the most likely age of husband for age of wife 35 years and most likely age of wife for age of husband 36 years.

Solution : ২৮৫ পৃষ্ঠার **Problem-30** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-20** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2019 No–12**

Calculate the two regression equations and the coefficient of correlation from the data given below :

Marks in Statistics (out of 50)	45	42	38	46	40
Marks in Mathematics (out of 50)	42	40	36	48	50

Solution : ২৯২ পৃষ্ঠার **Problem-34** নং প্রশ্নের উত্তর দেখুন।

০০০ **Problem-21** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2023 No–7**

Regression co-efficient of x on y is 3.5, regression co-efficient of y on x is 0.8. Is it true? Show the causes.

Solution : ২৯৪ পৃষ্ঠার **Problem-35** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

০০ **Problem-22** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2014 No–8**

From the following regression equations calculate the co-efficient of correlation :

$x = 5{\cdot}28 + 0{\cdot}59y$; $y = 1{\cdot}34x - 5{\cdot}40$.

Solution : ২৯৪ পৃষ্ঠার **Problem-36** নং প্রশ্নের উত্তর দেখুন।

☑ গড়, পরিমিত ব্যবধান, ভেদাংক, সংশ্লেষাংক দেওয়া থাকলে নির্ভরণ সমীকরণ নির্ণয়

ooo **Problem-23** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–16; (Acc.) 2nd 2016 No–14; (Marketing) 2nd 2017 No–12; (Mgt) 3rd 2016 No–14; 2018 No–14**

You are given below the following information about advertisement and sales :

	Advertisement Exp (x) Tk. lakh	**Sales (y) Tk. crores**
Mean	20	120
SD (σ)	5	25

Correlation co-efficient + 0.80

(i) Calculate two regression equations.

(ii) Find the likely sales when advertisement expenditure (x) is Tk. 25 lakh.

(iii) What should be the advertisement budget if the company wants to achieve sales target of Tk. 150 cores?

Solution : ২৯৭ পৃষ্ঠার **Problem-39** নং প্রশ্নের উত্তর দেখুন।

০০ **Problem-24** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2015 No–15**

You are given the following data above the sales and advertisement expenditure of a firm :-

	Sales (in lakh Tk.)	Advertising expenditure (in lakh Tk.)
Artihmetic Mean	50	10
Variance	100	4

Co-efficient of correlation = 0.9

(i) Estimate the likely sales for a proposed advertising expenditure of Tk. 13.5 lakh.

(ii) Estimate the likely advertising budget if the company wants to achieve a sales target of Tk. 70 lakh and also comments the result.

Solution : ২৯৮ পৃষ্ঠার **Problem-40** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-25** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2017 No–15(b); D.U. BBA (Hon's) 2nd (Marketing) 2017 No–13**

Given the following data :

	Advertising expenditure (Lakh Taka)	Sales (Taka in crore)
Mean	35.00	82.00
Standard deviation	9.03	17.16
Correlation of co-efficient = 0.61		

(i) Calculate two regression equations.
(ii) Estimate the likely sales for an advertising expenditure of Tk. 40 lakh.
(iii) What would be the advertising expenditure for attaining sales target Tk. 90 crore?
(iv) Comments on the reliability of the estimates.

Solution : ৩০১ পৃষ্ঠার **Problem-42** নং প্রশ্নের উত্তর দেখুন।

☑ নির্ভরণ সমীকরণ হতে গড়, নির্ভরাংক ও সংশ্লেষাংক নির্ণয়

oo **Problem-26** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2017 No–14**

You are supplied with the following information :—

$6x - 5y = 20$

$2x - y = 16$

The standard deviation of variable x is 4.

Calculate :

(i) The average value of x and y;
(ii) The co-efficient of regressions;
(iii) co-efficient of correlation.

Solution : ৩০৩ পৃষ্ঠার **Problem-43** নং প্রশ্নের উত্তর দেখুন।

☑ নির্ভরণ রেখা অংকনের অংক

oo **Problem-27** ***Year Question & Solution*** ⊙ **D.U. BBA (Hon's) 3rd (Mgt) 2016 No–11**

From the following information draw two regression lines:

X	70	85	92	105	190	240
Y	760	1225	1370	1750	1585	2290

If x = 300, what should be the value of y?

Solution : ৩০৪ পৃষ্ঠার **Problem-44** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-28** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2024 No–14**

From the following information calculate two regression equations and draw the regression line :

Advertisement (Tk. in lac)	40	45	48	50	55	60	65
Sales (Tk. in crore)	12	14	15	17	18	20	21

Solution : ৫৬৯ পৃষ্ঠার **Question No-14** নং প্রশ্নের উত্তর দেখুন।

Chapter 08 Forecasting and Time Series Analysis

(পূর্বাভাস ও কালিন সারি বিশ্লেষণ)

Part-A : Brief Questions :

01. What is time series analysis? [২০২৪ সালে যা এসেছিল]

Or, What do you mean by time series?

[Test Exam-Eden (Mgt), Dhaka Commerce, Govt. Bangla, Habibullahbaher, New Model; N.U. 2016, 2020, 2024 (Mgt.)]

Or, What is time series?

Ans : *When data are arranged in against of time, it is called the time series.*

02. What is the elaboration of ANOVA? [২০২৪ সালে যা এসেছিল] [N.U. 2017, 2024 (Mgt. 3rd year)]

Ans : *ANOVA = Analysis of Variance.*

✪ 03. **What is secular trend?**

Ans : A type of variation in a time series, the value of the variable tending to increase or decrease over a long period of time.

✪✪✪ 04 **Write down the components of time series.** 99%

[N.U. 2012 (old); 2015, 2017, 2023 (Mgt. 3rd year)]

Ans : *The components of time series are*

(i) Trend (ii) Seasonal variation

(iii) Cyclical variation

(iv) Irregular variation..

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 05. **What is business forecasting?** 99% *[N.U. 2015]*

Ans : Business forecasting involves predicting of future business condition of based on past data.

✪✪✪ 06. **What are the method of determining trend.**

Ans : There are 4 methods for determining trend-

(i) Free hand method;

(ii) Semi average method.

(iii) Moving average method;

(iv) Least square method.

✪✪ 07. **Define seasonal variation.**

Ans : Patterns of change in a time series within a year is called seasonal variation.

✪✪ 08. **Define cyclical variation.**

Ans : The periodic movements in a time series where the period is more than one year are called cyclic variation.

✪✪✪ 09. **Define irregular variation.**

Ans : A condition in a time series in which the value of variable is completely unpredictable.

✪✪ 10. **Write the methods of estimating trended?**

Ans : (i) Mean forecast

(ii) Naive forecast

(iii) Linear trend forecast

(iv) Non linear trend forecast.

✪✪✪ 11. **Give the examples of cyclic variation?**

Ans : Example : The production of jute in Bangladesh which is influenced by weather, Socio-economic condition, farmer's personal view etc.

✪✪✪ 12. **The additive model of time series is expressed as?**

Ans : Detrending = $y - \hat{y}$

✪✪✪ 13. **What is cyclical variation/fluctuation?**

Ans : A type of variation in a time series, in which the value of the variable fluctuates above and below a secular trend line.

✪✪✪ 14. **What is modified mean?**

Ans : A statistical method used in time-series analysis to discard the highest and lowest values when computing a mean.

✪✪✪ 15. **What is residual method?**

Ans : A method of describing the cyclical component of a time series. It assumes that most of the variation in the series not explained by the secular trend is cyclical variation.

✪✪✪ 16 **What is least-square method?** 99%

[N.U. (3rd)-2018 (Mgt.)]

Ans : The method of least squares is a standard approach in regression analysis to approximate the solution at over determined systems i.e, sets of equations.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ 17. **What is meant by H_O and H_A?**

Ans : H_O means null hypothesis (H Subscript o) H_A means Alternative Hypothesis (H Subscript 1/A) *[N.U. 2012 (old)]*

✪✪✪ 18. **What is meant by deseasonalization?**

Ans : A statistical process used to remove the effects of seasonality from a time series is known as deseasonalization.

✪✪ 19. **Define relative cyclical residual.**

Ans : Relative cyclical residual is a measure of cyclical variation. It uses the percentage deviation from the trend for each value in the series.

✪✪ 20. **What is coding?**

Ans : A method of converting traditional measures of time to a form that simplifies computation is called coding (often called translating).

✪✪✪ 21. **What is second degree equation?**

Ans : A mathematical form used to describe a parabolic curve that may be used in time series trend analysis.

✪✪✪ (22) **What is forecasting?** 99%

[N.U. Acc 3rd-2019; (2nd)-2015 (Marketing); D.U. (2nd)-2017 (Marketing)]

Ans : Forecasting in statistics refers to the process of estimating or predicting future values of a variable based on historical data and statistical methods.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

Part-B & C : Short & Broad Questions :

✪ 01. What is meant by secular trend? How is it measured? *[N.U. (Old) 2008, 2006]*

[উ: ব্যবস্থাপনা ইন্সিগ্নিয়া (২য় খণ্ড) বইয়ের ৩০৯ পৃষ্ঠার ৮.০৭ নং]

✪✪✪ 02. Explain the criteria of a good forecasting method. [উ : ৩০৮ পৃষ্ঠার ৮.০৫ নং] *[N.U. (Old) 2011; D.U- 2nd 2016]*

✪✪✪ (03) **What is time series? Describe the component of time series.** 99% [উ : ৩০৭+৩০৮ পৃষ্ঠার ৮.০৩ + ৮.০৪ নং]

[Test Exam-Tolaram, Dinajpur College; N.U. (Old)2005, 2007, 2009 ; D.U. (3rd)-2016 (Mgt.)]

Or, Discuss the elements of a time series.

[রকেট স্পেশাল সাজেশন্স: ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

✪✪✪ (04) **What is the necessity of Time series analysis?** 99% [উ : ৩০৯ পৃষ্ঠার ৮.০৬ নং] *[N.U. Mgt-2019]*

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 03 & 04 No. Questions in are most important for the Examinee of 2025.*** ***..... Editor***

Easy Math Basic Problem : Category-B & C

✪✪ **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2021 No–09**

From the following data, calculate trend value using 3-yearly moving average :

Year	2015	2016	2017	2018	2019	2020	2021
Production	412	440	448	454	470	483	495

Solution : ব্যবস্থাপনা ইন্সিগ্নিয়া (২য় খণ্ড) বইয়ের ৩১২ পৃষ্ঠার **Problem-02** নং প্রশ্নের উত্তর দেখুন।

✪✪✪ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2023 No–9**

Calculate 4-yearly moving average of the students of 3rd year BBA class;

Year	2016	2017	2018	2019	2020	2021	2022
No. of students	45	60	50	70	65	80	90

Solution : ৩১৩ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

✪✪✪ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2019 No– 09**

Calculate 4 yearly moving average of the students of a business studies class :

Year	2012	2013	2014	2015	2016	2017	2018
No. of students	35	40	55	65	70	65	30

Solution : ৩১৪ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Old) (Hon's) 2nd 2011 No–9(b); 3rd (Mgt.) 2020 No–07; 2021 No–14**

From the following data, calculate trend values using 3 yearly moving average.

Year	2015	2016	2017	2018	2019	2020	2021
Production	412	438	446	454	470	483	490

Also plot the original and trend values on the same graph paper.

Solution : ৩১৪ পৃষ্ঠার **Problem-09** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2019 No–16**

Fit a trend line by the method of least squares to the following data :

Year	2005	2006	2007	2008	2009	2010	2011	2012
Sales (Tk. crore)	12	40	60	70	75	60	85	95

Solution : ৩১৬ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2023 No–14**

Fit a trend line by using least square method from the following data and estimate the sales for the year 2025.

Year	2017	2018	2019	2020	2021	2022	2023	2024
Sales (In lakh Tk.)	35	38	62	71	70	62	89	97

Solution : ৩১৯ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

[রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

oo **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2017 No–15**

(a) Determine the trend free time series applying the method of least square from the following information :

Year	2010	2011	2012	2013	2014	2015
Income (Tk. in lac)	50	60	45	40	30	60

(b) Determine the monthly income.

Solution : ৩২০ পৃষ্ঠার **Problem-14** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-]

xx **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2024 No–16**

The productions of a rice mill are given below :

Year	2018	2019	2020	2021	2022	2023	2024
Production (In thousand tons)	72	90	98	96	99	102	110

(i) Fit a straight line by the least square method and tabulated the trend value;

(ii) What is the monthly increase or decrease in the production of rice?

Solution : ৫৭২ পৃষ্ঠার **Question No-16** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–17; (Marketing) 2015 No–13; 3rd (Mgt.) 2016 No–16; 2022 No– 13**

Fit a trend line by the method of least squares to the following data :

Year	2014	2015	2016	2017	2018	2019	2020	2021
Sales	38	40	60	70	75	60	85	95

Plot the given data on a graph paper and estimate the sales for the year 2022.

Solution : ৩২২ পৃষ্ঠার **Problem-16** নং প্রশ্নের উত্তর দেখুন।

০০০ (৯৯%) [রকেট স্পেশাল সাজেশন্স : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। –সম্পাদক]

ooo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2020 No–14**

Fit a trend line by the method of least squares to the following data :

Year	2013	2014	2015	2016	2017	2018	2019	2020
Sale (In Tk. lac)	38	40	65	72	69	60	87	95

Plot the given data on a grap paper. Showing also the trend line, estimate the sales for year 2021.

Solution : ৩২৩ পৃষ্ঠার **Problem-17** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-11** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Marketing) 2017 No–13; 3rd (Mgt.) 2019 No–15**

Given below are the figures of production (in thousand tones) of a factory.

Year	1994	1995	1996	1997	1998	1999	2000
Production	80	90	92	83	94	99	92

(i) Show the trend equation.
(ii) Calculate the trend values and draw trend line.
(iii) Calculate trend-free data.
(iv) Calculate the likely production of 2001.
(v) Calculate the monthly increase or decrease.

Solution : ৩২৫ পৃষ্ঠার **Problem-18** নং প্রশ্নের উত্তর দেখুন।

Chapter 09 Index Number

(সূচক সংখ্যা)

Part-A : Brief Questions & Solutions :

০০০ (01) **What is index number?** 99% *[Test Exam-Eden (Acc.) Govt. Bangla, Habibullahbaher College; N.U. (Old) 2011; (2nd)-2018 (Finance); 2021 (Mgt. 3rd)]*

Ans : Index number may be defined as the ratio that measures how much a variable changes over time.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

০০০ 02. **What is price index?** *[Test Exam-Eden (Mgt), Govt. Bangla College; N.U. (Old) 2013; (Acc)-2018]*

Ans : The price index measures the relative change in the retail or wholesale price level of any particular commodity or a group of commodities.

০০০ (03) **What is industrial index?** 99% *[N.U. 2022 (Hon's) 3rd]*

Ans : The industrial index is a business cycle indicator which measures monthly changes in the price adjusted output of industry.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। –সম্পাদক]

ooo 04. **What is value index?** 99% *[N.U. (3rd)-2017(Mgt.)]*

Ans : A value index measures general changes in the total value of some variable over the time.

oo 05. **What is laspeyres Index.**

Ans : Laspeyres Index $= \frac{P_1q_0}{P_0q_0} \times 100$

ooo 06. **Write the formula Paasche's price Index.**

Ans : Paasche's price Index.

$$P_{0n} = \frac{\Sigma p_n q_n}{\Sigma p_0 q_n} \times 100$$

ooo 07. **What is family budget method.**

Ans : When the consumer price index is computed on the family budgets of a large number of people, is called family budget method.

ooo 08. **What is weighted aggregate index?**

Ans : Using all the values considered this index assigns weights to these values.

ooo 09. **What is unweighted aggregates index?**

Ans : Uses all the values considered and assigns equal importance to each of these values.

o 10. **What is chain index.**

Ans : When several bases are considered instead of a fixed base, it is called chain index.

ooo 11. **State the meaning 'TRT' & 'F RT'**

Ans : TRT means Time reversal test and FRT means Factor reversal test.

12. What is TRT? [২০২৪ সালে যা এসেছিল]

[N.U. 2024 (Mgt. 3rd)]

Ans : *Time reversal test is a test to determine whether a given method will work both ways in time, forward and backward.*

ooo 13. **What is index number?**

Ans : To weight an aggregates index, the Passche method uses as weights the quantities consumed during the current period.

oo 14. **Under what condition time reversal test and factor reversal test is satisfied?**

Ans : Time reversal Test is satisfied when

${}_0P_n \times {}_nP_0 = 1$ *[N.U. (Old) 2012]*

Factor reversal test is satisfied when.

$${}_0P_n \times {}_0q_n = {}_0V_n = \frac{\sum p_n q_n}{\sum p_0 q_0}$$

ooo 15. **What is the ideal index?** 99% *[N.U. -2015]*

Ans : The index number which give the best representative value and satisfies time reversal test (IRT) and factor reversal test (FRT) is called an ideal index.

ooo 16 **Which index number is ideal?** 99%

[N.U. (Old) 2013; Acc 3rd-2019; 2023 (Mgt.)]

Ans : Fisher Index number is ideal.

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্বসহকারে পড়তে হবে। -সম্পাদক]

ooo 17. **What is the formula of ideal index?** *[N.U. -2015]*

Ans : Formula of ideal index (fisher's index)

Price index number

$$= \sqrt{\frac{\sum P_1 q_0}{\sum P_0 q_0} \times \frac{\sum P_1 q_1}{\sum P_0 q_1}} \times 100$$

Quantity index number

$$= \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}} \times 100$$

Part-B & C : Short & Broad Questions :

ooo 01. **Discuss the uses of index numbers.** 99% [উ ব্যবসায় পরিসংখ্যান (২য় খণ্ড) বইয়ের ৩২৯ পৃষ্ঠার ৯.০৮ নং]
Or, Discuss the importance of index number. *[N.U. (Old) 2004, 2010, (Acc-2nd) 2015; Mgt-2018]*
Or, Why index number is called 'Economic Barometer'?

oo 02. Prove that fisher's index can satisfy both TRT and FRT. [উ : ৩২৭ পৃষ্ঠার ৯.০৪ নং] *[N.U. (3rd)-2017 (Mgt.)]*

o 03. Distinguish between Price index number and quantity index number. [উ : ৩৩২ পৃষ্ঠার ৯.১৬ নং]

oo 04. What are the purposes of constructing index number? [উ : ৩২৯ পৃষ্ঠার ৯.০৬ নং] *[N.U. (3rd)-2018 (Mgt.)]*

ooo 05 **Distinguish between price index and cost of living index.** 99% [উ : ৩৩১ পৃষ্ঠার ৯.১৫ নং] *[N.U. 2017, 2020 (Mgt.)]*

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

oo 06. What is index number? What are the different types of index numbers? [উ : ৩২৭+৩৩০ পৃষ্ঠার ৯.০১ + ৯.১০ নং] *[Test Exam : Titumir, M.M, Al-Hera, Comilla, Munsur Ali College; N.U. (Old) -2004, 2009, 2018 (Mgt.)]*
Or, Explain index number. Discuss about various types of index number.

07 **Which index number is called ideal and why?** 99% [উ : ৩২৭ পৃষ্ঠার ৯.০৩ নং] *[N.U. (3rd)-2018 (Mgt.)]*

Or, Why is fisher index called ideal?

Or, What do you mean by fisher ideal index?

[রকেট স্পেশাল সাজেশন্স: উপরিউক্ত ৯৯% প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

Help Line ***Dear Students, In above Suggestions Part-B and Part-C : 01, 05 & 07 No. Questions in are most important for the Examinee of 2025. ..... Editor***

Easy Math Basic Problem : Category-B & C

☑ সূচক সংখ্যার মান নির্ণয়

oo **Problem-01** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2018 No–8**

From the following data calculate Index Number by Simple Aggregate Method :

Commodity	A	B	C	D
Price in 2017	162	256	257	132
Price in 2018	171	164	189	145

Solution : ব্যবস্থাপনা সাজেশন্স (২য় খণ্ড) বইয়ের ৩৩৭ পৃষ্ঠার **Problem-01** নং প্রশ্নের উত্তর দেখুন।

[২০২৪ সালের পরীক্ষায় যা এসেছিল-

☒☒ **Problem-02** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Acc.) 2015 No–15; 3rd (Mgt) 2024 No–17**

From the given data construct index number of prices for 2024 with 2021 as base. By using–

(i) Laspeyre's method

(ii) Paasche's method and

(iii) Fisher's method :

Commodity	2021		2024	
	Price per unit	Expenditure (Tk.)	Price per unit	Expenditure (Tk.)
A	2	10	4	16
B	3	12	6	18
C	1	8	2	14
D	4	20	8	32

Solution : ৩৩৭ পৃষ্ঠার **Problem-03** নং প্রশ্নের উত্তর দেখুন।

☑ অংকের সাহায্যে সময় উল্টানো এবং উপাদান উল্টানো পরীক্ষা

oo **Problem-03** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Accounting) 2015 No–9; D.U. (Accounting) 2017 No–7**

From the following data verify the "Time Reversal Test" and "Factor Reversal Test":

$\Sigma p_1 q_0 = 150$; $\Sigma p_1 q_1 = 200$, $\Sigma p_0 q_1 = 100$; $\Sigma p_0 q_0 = 50$.

Solution : ৩৪১ পৃষ্ঠার **Problem-04** নং প্রশ্নের উত্তর দেখুন।

☑ সূচক সংখ্যার মান নির্ণয় এবং সময় উল্টানো এবং উপাদান উল্টানো পরীক্ষা

ooo **Problem-04** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2012 No–13; 2nd (Finance) 2017 No–17(b); 3rd (Mgt) 2022 No–12**

From the help of following data shown that Fisher's ideal index number formula satisfies the time reversal test and factor reversal test :

Commodity	2021		2022	
	Price (Tk.)	Quantity (unit)	Price (Tk.)	Quantity (unit)
A	56	71	50	26
B	32	107	30	83
C	41	62	28	48

Solution : ৩৪২ পৃষ্ঠার **Problem-05** নং প্রশ্নের উত্তর দেখুন।

০৫৯ (৯৯%) [রকেট স্পেশাল সাজেশন : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

ooo **Problem-05** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Old) 2014 No–16; (Accounting) 2017 No–12; (Marketing) 2018 No–15; 3rd (Mgt) 2020 No–15**

Compute Fisher's Index Number from the following data and verify the time reversal test and factor reversal test:

Commodity	2019		2020	
	Price	Quantity	Price	Quantity
A	10	12	12	15
B	7	15	5	20
C	5	24	9	20
D	16	5	14	5

Solution : ৩৪৩ পৃষ্ঠার **Problem-06** নং প্রশ্নের উত্তর দেখুন।

ooo **Problem-06** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2023 No–17**

Determine Fisher's index number by using the following data and verify the time reversal test and factor reversal test :

Commodity	2023		2024	
	Price	Quantity	Price	Quantity
A	8	10	10	12
B	8	14	5	17
C	5	20	8	18
D	17	6	15	6

Solution : ৩৪৫ পৃষ্ঠার **Problem-07** নং প্রশ্নের উত্তর দেখুন।

০৫৯ (৯৯%) [রকেট স্পেশাল সাজেশন : উপরিউক্ত প্রশ্নটি ১০০% গুরুত্ব সহকারে পড়তে হবে। -সম্পাদক]

০৭ **Problem-07** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2016 No–17**

From the following information calculate, Fisher's ideal index number and also comment:—

Products	Base Year		Current Year	
	Price in Tk.	Quantity	Price in Tk.	Quantity
A	6	50	10	56
B	2	100	2	120
C	4	60	6	60
D	10	30	12	24
E	8	40	12	36

Solution : ৩৪৭ পৃষ্ঠার **Problem-08** নং প্রশ্নের উত্তর দেখুন।

☑ জীবন যাত্রার ব্যয় সূচক সংখ্যা

ooo **Problem-08** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Acc.) 2016 No–16; 3rd (Mgt) 2021 No–16**

Calculate cost of living index from the following data and discuss the change of living cost with respect of 2011 to 2012.

	Food	House rent	Cloth	Fuel	Others
Cost	30%	15%	20%	10%	25%
Price (2011)	100	20	70	20	40
Price (2012)	90	20	60	15	55

Solution : ৩৫০ পৃষ্ঠার **Problem-10** নং প্রশ্নের উত্তর দেখুন।

oo **Problem-09** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 3rd (Mgt) 2018 No–16(b); (MBA Preli) (Mgt) 2017 No–8(c)**

Calculate the cost of living index from the following data :

Product	Per unit value(Tk.)		Total expenses in 2013 (Tk.)
	2013	2014	
Rice	320	340	11,600
Wheat	250	300	6,500
Pulse	500	550	1,500
Sugar	540	600	1,080
Milk	1,200	1,400	3,000

Solution : ৩৫২ পৃষ্ঠার **Problem-13** নং প্রশ্নের উত্তর দেখুন।

☑ প্রকৃত আয় বা Real Income এর অংক

oo **Problem-10** ***Year Question & Solution*** ⊙ **N.U. BBA (Hon's) 2nd (Finance) 2019 No–14**

From the following information test how is real income changed by using cost of living index :

Year	2012	2013	2014	2015	2016
Monthly Income	5000	6000	7000	8000	9000
Living Index	150	200	250	300	350

Solution : ৩৫৪ পৃষ্ঠার **Problem-15** নং প্রশ্নের উত্তর দেখুন।

গ্রন্থ সহায়ক : উপরিউক্ত সাজেশন্সের পূর্ণাঙ্গ উত্তরপত্র পেতে.....

ব্যতিক্রম ব্যবস্থাপনা ইজিপ্রেস (২য় খণ্ড) বইটি সংগ্রহ করুন।

জাতীয় বিশ্ববিদ্যালয়ের প্রশ্নাবলি ও উত্তরপত্র

ব্যবসায় পরিসংখ্যান: বিষয় কোড : ২৩২৬০৩

National University

[BBA (Hon's) 3rd Year Exam-2019 (Will be held-05/02/2020)]

(Department of Management)

Subject : Business Statistics (In English)

Subject Code : 232603

Time : 4 hours **Full marks :** 80

[**N.B.** Different parts of a question answered chronologically.]

Part-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is statistics?

Ans : Statistic is the science of decision making in the field of uncertainty.

(b) What is ogive?

Ans : Ogive is a graph of cumulative frequency distribution.

(c) What is modal class?

Ans : The class which belongs highest frequency is called modal class. No extra column is required for calculating mode.

(d) What is the range between 1 and –1?

Ans : Range = Highest value – Lowest value.

$= 1 - (-1)$

$= 1 + 1$

$= 2$

(e) What are the minimum and maximum values of probability?

Ans : Minimum value 0 and maximum value 1.

(f) What is event?

Ans : One or more of the possible outcomes of doing something, or one of the possible outcomes from conducting an experiment.

(g) What is population?

Ans : A population is the collection of all possible objects having certain characteristic under a study.

(h) Define type-1 error.

Ans : Type 1 error is the rejecting a null hypothesis when it is true.

(i) What is meant by r = 1, r = –1 and r = 0?

Ans : The value of r = 1, Means the perfect positive correlation.

The value of r = –1, Means called perfect Negative correlation.

The value of r = 0 Means called Zero (0) correlation.

(j) What is co-efficient of correlation?

Ans : Co-efficient of correlation is the measure of the strength of the linear relationship between two variables.

(k) What is forecasting?

Ans : Busies formatives is Contained with predication of future Condition of basemen of the basis of past data.

(l) Which index number is ideal?

Ans : Fisher index number is ideal.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. Describe the limitations of statistics.
[Ans : বিবিএ ব্যবস্থাপনা ইন্সপেস (২য় খণ্ড) বইয়ের Page No. 5 Ques. No. 1.06]

3. What are the methods of collecting secondary data? *[Ans : Page No. 7 Ques. No 1.14]*

4. Represent the following data by histogram and calculate mode from histogram :

Yearly Income	10—20	20—30	30—40	40—50	50—60	60—70
No. of Family	125	350	170	60	25	10

Solution *Page No-21; Problem No-02.*

5. Two samples of size 100 and 150 respectively have means 50 and 60 and standard deviations 5 and 6. Find the mean and standard deviation of the combined sample of size 250.
Solution *Page No-136; Problem No-18.*

6. What is the necessity of Time Series Analysis? *[Ans : Page No. 309 Ques. No 8.06]*

7. In a moderately skewed frequency distribution, the mean is Tk. 15 and the median is Tk. 14. If the co-efficient of variation is 30%, find the Pearsonian co-efficient of skewness.
Solution *Page No-149; Problem No-32.*

8. Show the differences between Co-rrelation and Regression. *[Ans : Page No. 245 Ques. No 7.21]*

9. Calculate 4 yearly moving average of the students of a business studies class :

Year	2012	2013	2014	2015	2016	2017	2018
No. of students	35	40	55	65	70	65	30

Solution *Page No-313; Problem No-08.*

Group-C

[Answer any **five** questions.] $5 \times 10 = 50$

10. Find the mean, median and mode from the following information :

Marks	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69	70 – 79	80 – 89	90 – 99
No. of students	2	12	15	20	18	10	6	8

Solution *Page No. 74; Problem No. 08*

11. Prices of a particular commodity in five years in two cities are given below :

Price in city-A	Price in city-B
20	10
22	20
19	18
23	12
16	15

From the above data find the city which had more stable prices and why?
Solution *Page No-125; Problem No-12.*

12. Find the co-efficient of correlation (r) between X and Y from the following data:

X	3	6	5	4	4	6	7	5
Y	3	2	3	5	3	6	6	4

Also calculate the regression of Y on X and predict the average value of Y when X is 9.
Solution *Page No-249; Problem No-04.*

13. (a) Define population, sample and sampling with example. *[Ans : Page No. 200+201 Ques. No 5.01+5.05+5.07]*
(b) Explain the advantages of sample survey compared to population survey. *[Ans : Page No. 200 Ques. No 5.04]*

14. About 20% items of a production are defective in a certain factory. 10 products are packed in a packet. Among the 2000 packets find the probability of getting packet containing :
(i) no defective product.
(ii) at least one product is defective
(iii) at best one product is defective.
Solution *Page No-189; Problem No-25.*

15. Fit a straight line trend by the method of least squares to the following data and also show on graph paper :

Year	2012	2013	2014	2015	2016	2017	2018
Production in '000	80	90	92	83	94	99	92

Solution ***Page No-324; Problem No-18.***

16. (a) What do you mean by Bi-variate frequency distribution? *[Ans : Page No. 16 Ques. No 1.46]*

(b) The following data represent the marks in Business Mathematics (X) and Business Statistics (Y) of 13 students. Prepare a Bi-variate frequency table from the following data :

X	32	15	21	27	58	37	47	58	46	50	40	54	62
Y	75	48	54	70	92	53	74	85	65	62	73	77	40

Solution ***Page No-47; Problem No-24.***

17. What is Kurtosis? Describe the various types of Kurtosis with diagram. *[Ans : Page No. 110 Ques. No 3.40+3.41]*

National University

[BBA (Hon's) 3rd Year Exam-2020 (Held on-16/03/2022)]

(Department of Management)

Subject : Business Statistics (In English)

Subject Code : 232603

Time : 4 hours **Full marks :** 80

[N.B. Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is pie-chart?

Ans : A pie-chart is formed by marking off a portion of the pie corresponding to each characteristic being displayed.

(b) What is primary data?

Ans : primary data that have been collected from the origin of the fields is called primery data.

(c) What is arithmetic mean?

Ans : An Arithmetic Mean is a numerical value obtained by dividing the Sun of values of all the observations by the total number of observations.

(d) Calculate range of the followings : 2, 3, 4, – 2, – 4.

Ans : Range = Highest value – Lowest value

$= 4 - (-2)$

$= 4 + 2$

$= 6$

(e) How many 'types of skewness?

Ans : Skewness is of two types as : (i) Positive Skewness, (ii) Negative Skewness.

(f) What is rank correlation?

Ans : The Co-efficient of correlation between two sets of rank is called rank correlation.

(g) Define standard deviation.

Ans : The standard deviation is the positive square root of the mean of the squared deviations from their mean of a set of value.

(h) What is time series?

Ans : When data are arranged in against of time, it is called the time series.

(i) What is scatter diagram?

Ans : The simplest device for studying correlation in two variables is a special type of at chart called scatter diagram.

(j) What is histogram?

Ans : Histogram is a Suitable graph for representing the frequency distribution of a continuous series.

(k) Define null hypothesis.

Ans : The hypothesis about a Population Parameter we wish to test is called a null hypothesis.

(l) What is critical value?

Ans : The value of the standard statistic (Z or t) beyond which we reject the null hypothesis.

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. What is the role of statistics in managerial decision making?
[Ans : বেটাভিত্তিক ব্যবস্থাপনা স্টাডিগাইড (২য় খণ্ড) বইয়ের Page No. 14 Ques. No. 1.41]
3. Show the difference between primary data and secondary data. *[Ans : Page No. 6 Ques. No 1.12]*
4. Distinguish between price index and cost of living index. *[Ans : Page No. 331 Ques. No 9.15]*
5. The average marks of 39 students of a class are 50. The marks obtained by 40th student are 39 more than the average marks of all the 40 students. Find the mean marks of all the 40 students.
Solution *Page No-67; Problem No-01.*
6. What are the properties of a good measure of dispersion? *[Ans : Page No. 103 Ques. No 3.13]*
7. From the following data, calculate trend values using 3 yearly moving average :

Year	1981	1982	1983	1984	1985	1986	1987
Production	412	438	446	454	470	483	490

Solution *Page No-313; Problem No-09.*
8. From the following data calculate mean deviation :
16, 20, 22, 19, 18, 20, 30, 18, 20, 22. **Solution** *Page No-120; Problem No-08.*
9. Five cards are drawn from a packet of well-shuffled cards. Find the probability that 4 cards are aces.
Solution *Page No-183; Problem No-16.*

Group-C

[Answer any **five** questions.] $5 \times 10 = 50$

10. Calculate the mode from the following data :

Income in Tk.	500–1000	1000–1500	1500–2000	2000–2500	2500–3000
No. of persons	13	11	10	8	5

Solution *Page No. 70 Problem No. 05.*
11. Find out skewness of the following data and comment on the results : **Solution** *Page No-153; Problem No-36.*

Marks obtained	No. of students
40—50	8
50—60	12
60—70	20
70—80	25
80—90	15

12. Define dispersion. Discuss the different measures of dispersion. *[Ans : Page No. 101+104 Ques. No 3.01+3.14]*
13. Find out the co-efficient of correlation of the following data and comment on the result:

X (Income)	50	720	202	130	140	182
Y (Expenditure)	32	378	105	86	84	72

Solution *Page No-256; Problem No-08.*
14. Fit a trend line by the method of least squares to the following data :

Year	2013	2014	2015	2016	2017	2018	2019	2020
Sale (In Tk. lac)	38	40	65	72	69	60	87	95

Plot the given data on a grap paper. Showing also the trend line, estimate the sales for year 2021.
Solution *Page No-322; Problem No-17.*
15. Compute Fisher's Index Number from the following data and verify the time reversal test and factor reversal test:

Commodity	2019		2020	
	Price	Quantity	Price	Quantity
A	10	12	12	15
B	7	15	5	20
C	5	24	9	20
D	16	5	14	5

Solution *Page No-341; Problem No-06.*

16. Three unbiased coins are thrown at a time, construct a sample space. What is the probability of getting :
    (i) At best two heads;
    (ii) At least two heads. **Solution** *Page No-175; Problem No-04.*

17. Calculate the skewness and curtosis from the following information and comment on the result:

Population in family	2	3	4	5	6	7
Number of family	250	310	400	200	120	75

Solution *Page No-162; Problem No-44.*

National University

[BBA (Hon's) 3rd Year Exam-2021 (Held on-07/05/2023)]

(Department of Management)

Subject : Business Statistics (In English)

Subject Code : 232603

Time : 4 hours **Full marks :** 80

[**N.B.** Answer different questions of each group chronologically.]

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is business statistics?

Ans : Business statistics is a science assisting us to make decision under uncertainties based on some numerical and measurable scales.

(b) What is population?

Ans : Population refers to all subject possessing a common characteristic that is being studied.

(c) When AM, GM and HM are equal?.

Ans : If each value of a observation are equal then arithmetic mean, geometric mean and harmonic are equal.

(d) Which measure is suitable for measures of dispersion?

Ans : Standard deviation is suitable measures of dispersion.

(e) What is correlation?

Ans : Co-efficient of correlation is the measure of the strength of the linear relationship between two variables.

(f) Find the range between 1 and –1.

Ans : Range = Highest value – Lowest value. $= 1 - (-1) = 1 + 1 = 2$

(g) What is regression analysis?

Ans ; Regression Analysis is a mathematical measure of the average relationship between two or more variables in terms of original units of the data.

(h) What is sampling?

Ans : The methods for selecting sample from a population is known in sampling.

(i) Which test is known as small test?

Ans : T-test is known as small test. Anather name of t-test is student test.

(j) What is kurtosis?

Ans : The word 'kurtosis' comes from a Greek word meaning 'humped'. In statistics, it refers to the degree of flatness or peakedness in the region about the mode of a frequency curve.

(k) What is meant by r = + 1?

Ans : r = 1 means perfect positive correlation between x and y.

(l) What is index number?

Ans : Index number may be defined as the ratio that measures how much a variable changes over time.

Group-B

[Answer any **five** questions.] Mark— $4 \times 5 = 20$

2 Discuss the scope of business statistics.

[Ans : ব্যবস্থাপনা পরিসংখ্যান (২য় খণ্ড) বইয়ের **Page No. 4 Ques. No. 1.02]**

3. Describe the sources of secondary data. *[Ans : Page No. 7 Ques. No 1.14]*

4. Distinguish between absolute and relative measures of dispersion. *[Ans : Page No. 105 Ques. No 3.21]*

5. Draw a pie-chart for the following data :
Cost of materials Tk. 40,000, Cost of labour Tk. 30,000, Direct expenses of manufacture Tk. 10,000, Factory overhead expenses Tk. 15,000 and Miscellaneous expense Tk. 5,000.
Solution *Page No-41; Problem No-18.*

6. There are 5 red and 4 white balls in a bag. One ball is drawn from the bag. What is the probability that it is either red or white?
Solution *Page No-176; Problem No-06.*

7. Prove that $-1 \le r \le +1$. *[Ans : Page No. 243 Ques. No 7.17]*

8. Intelligence test on groups of boys and girls gives the following results.

	Mean	S.D	N
Girls	75	15	150
Boys	70	20	250

Is there a significant difference in the mean scores obtained by boys and girls?
Solution *Page No-223; Problem No-05.*

9. Two groups containing 200 and 300 observations. Their arithmetic mean is 15 and 20, standard deviation is 3 and 5 respectively. Calculate combined mean and standard deviation.
Solution *Page No-137; Problem No-19.*

Group-C

[Answer any **five** questions.] Marks — $10 \times 5 = 50$

10. The obtained marks of 30 candidates in an examination are given below :

40	30	35	55	68	72	60	52	48	62
45	47	33	57	75	87	56	49	45	39
40	45	57	59	62	65	72	73	85	90

(i) Prepare a frequency distribution table by using suitable class interval;
(ii) Construct a steam and leaf diagram. **Solution** *Page No. 44 Problem No. 21*

11. Find the missing frequencies from the incomplete table, when median is 45.

Class interval	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	Total
Frequency	5	15	?	30	25	?	5	120

Solution *Page No-81; Problem No-13.*

12. Particulars regarding income of two villages are given below :

	Village A	Village B
No. of people	600	500
Average income	175	186
Variance of income	100	81

Required :
(i) In which village is the variation in income greater?
(ii) What is the average income of the people of both the villages put together?
(iii) What is the combined standard deviation?
Solution *Page No-137; Problem No-20.*

13. From the following table, calculate regression equation of profit on capital and estimate profit when capital is 32 lakh. **Solution** *Page No-277; Problem No-26.*

Capital (lakh) :	5	10	15	20	25
Profit (lakh) :	3	4	8	12	18

14. From the following data, calculate trend values using 3 yearly moving average.

Year	2015	2016	2017	2018	2019	2020	2021
Production	412	438	446	454	470	483	490

Also plot the original and trend values on the same graph paper.
Solution *Page No-313; Problem No-09.*

15. In a random sample of 600 men taken from a big city in which 400 are found to be smokers. In another simple random, sample of 900 men taken from another city in which 450 are found smokers. Do the data indicate there is a significant difference in the habit of smoking in the two cities?

Solution *Page No-231; Problem No-15.*

16. Calculate cost of living index from the following data and discuss the change of living cost with respect of 2011 to 2012.

	Food	House rent	Cloth	Fuel	Others
Cost	30%	15%	20%	10%	25%
Price (2011)	100	20	70	20	40
Price (2012)	90	20	60	15	55

Solution *Page No-348; Problem No-10.*

17. From the following marks distribution, find Mean and Median marks.

Marks	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
No. of students	2	12	15	20	18	10	6	8

Solution *Page No-74; Problem No-08.*

National University

[BBA (Hon's) 3rd Year; Exam-2022 (Held in-28.04.2024)]

(Department of Management)

Subject Code : 232603

Subject Name : Business Statistics (In English)

Time : 4 hours; **Full Marks :** 80

[N. B. : Answer the different questions from each group chronologically.)

Group-A

1. Answer any **ten** questions. $1 \times 10 = 10$

(a) What is meant by descriptive statistics?

Ans : Descriptive statistics are the methods that help to collect, summarize, present and analyse a set of data.

(b) What is mode?

Ans : Mode is the value which occurs the greatest number of frequency in a series. Mode is the most fashionable or typical value of a distribution, because it is repeated the highest number of times in the series.

(c) Write the formula of variance.

Ans : Variance $= \frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2 = \left(\frac{\sum x}{N}\right)^2 - \left(\frac{\sum fx}{N}\right)^2$

(d) What is meant by event?

Ans : One or more of the possible outcomes of doing something, or one of the possible outcomes from conducting an experiment.

(e) What is normal distribution?

Ans : A distribution of a continuous random variable with a sing peaked bell shaped curve is called a normal distribution.

(f) What is cluster sampling?

Ans : Cluster sampling is a method of random Sampling in which the Population is divided into groups.

(g) What is meant by 'Type 1 Error'?

Ans : Type 1 error is the rejecting a null hypothesis when it is true.

(h) Define independent variable.

Ans : The know variable as variables in regression analysis is called independent variables.

(i) What is industrial index?

Ans : The industrial index is a business cycle indicator which measures monthly changes in the price adjusted output of industry.

(j) **What is critical value?**

Ans : The value of the standard statistic (Z or t) beyond which we reject the null hypothesis.

(k) **What is primary data?**

Ans : primary data that have been collected from the origin of the fields is called primary data.

(l) **Find the Geometric mean of the series 2, 4 and 8.**

Ans : Let, $x_1 = 2, x_2 = 4, x_3 = 8$

$$\therefore \text{Geometric mean} = (x_1 . x_2 . x_3)^{\frac{1}{3}}$$

$$= (2 \times 4 \times 8)^{\frac{1}{3}}$$

$$= (64)^{\frac{1}{3}}$$

$$= (4^3)^{\frac{1}{3}}$$

$$= 4$$

Group-B

[Answer any **five** questions.] $4 \times 5 = 20$

2. What is the role of statistics in managerial decision making?

[Ans ব্যবস্থাপনা পরিসংখ্যান (২য় খণ্ড) বইয়ের **Page No. 14 Ques. No. 1.41]**

3. Draw a frequency polygon from the following data :

Class interval	Frequency
20 – 29	50
30 – 39	20
40 – 49	80
50 – 59	15
60 – 69	35

Solution ***Page No-28; Problem No-08.***

4. Write the properties of a good measure of dispersion. ***[Ans : Page No. 103 Ques. No 3.13]***

5. Show the differences between mean deviation and standard deviation. ***[Ans : Page No. 105 Ques. No 3.22]***

6. From the following data calculate the Rank Co-efficient of correlation :

x	4	6	5	8	10
y	2	1	3	5	8

Solution ***Page No-273; Problem No-23.***

7. The average marks of 49 students of a class is 60. The marks obtained by 50th student is 49 more than the average marks of all the 50 students. Find the mean marks of all the 50 students.

Solution ***Page No-67; Problem No-02.***

8. From the following data calculae Karl Pearson's skewness and make a comment :

Solution ***Page No-149; Problem No-30.***

Mean	0.2314
Median.	0.25
Standard deviation	1.56

9. Give the following information's find standard deviation :

$N = 10, \sum x = 60, \sum x^2 = 1000.$

Solution ***Page No-115; Problem No-05.***

Group-C

[Answer any **five** questions.] Marks — 10 × 5 = 50

10. The following table gives the height (in inches) of 100 students of a class. Compute mean, mode and median of the height. Also comment about the nature of distribution :

Height (Inches)	60 – 62	62 – 64	64 – 66	66 – 68	68 – 70	70 – 72
No. of students	5	18	42	20	8	7

Solution *Page No. 73 Problem No. 07*

11. 3 cards are drawn at random from a pack of 52 cards. Find the probability of getting :
    (i) Two kings
    (ii) Two spades
    (iii) One spade, One club and one diamond and
    (iv) Two face cards. **Solution** *Page No-185; Problem No-20.*

12. From the help of following data shown that Fisher's ideal index number formula satisfies the time reversal test and factor reversal test :

Solution *Page No-340; Problem No-05.*

Commodity	2021		2022	
	Price (Tk.)	Quantity (unit)	Price (Tk.)	Quantity (unit)
A	56	71	50	26
B	32	107	30	83
C	41	62	28	48

13. Fit a trend line by the method of least squares to the following data :

Solution *Page No-321; Problem No-16.*

Year	2014	2015	2016	2017	2018	2019	2020	2021
Sales	38	40	60	70	75	60	85	95

Plot the given data on a graph paper and estimate the sales for the year 2022.

14. (a) Define population, sample and sampling with example. *[Ans : Page No. 200+201 Ques. No 5.01+5.05+5.06]*
    (b) Explain the advantages of sample survey compared to population survey. *[Ans : Page No. 200 Ques. No 5.04]*

15. Below is given the frequency distribution of weekly wages of 100 workers in a factory :

Monthly wages (Tk.)	No. of workers
3000 – 3500	3
3500 – 4000	5
4000 – 4500	12
4500 – 5000	23
5000 – 5500	31
5500 – 6000	10
6000 – 6500	8
6500 – 7000	5
7000 – 7500	3

Draw the ogive for the distribution and use it to determine the median of a worker.

Solution *Page No-32; Problem No-11.*

16. Price of shares (In Tk.) of a company on the different days in a month were found to be :
    66, 65, 69, 70, 69, 71, 70, 63, 64, 68
    Test whether the mean price of the shares in the month is Tk. 65.

Solution *Page No-234; Problem No-20.*

17. Calculate the Skewness and Curtosis from the following information and comment on the result :

Population in family	2	3	4	5	6	7
Number of family	250	310	400	200	120	75

Solution *Page No-162; Problem No-44.*

National University

[BBA (Hon's) 3rd Year; Exam-2023 (Held on-27/04/2025)]

(Department of Management)

Subject Code : 232603

Subject : Business Statistics (In English)

Time : 4 hours **Full marks :** 80

[N.B. Answer different questions of each group chronologically.]

Group-A

1. Answer any ten questions. $1 \times 10 = 10$

(a) Who is considered as the father of modern statistics?

Ans : R.A. Fisher.

(b) The mean of a symmetrical distribution is 40.75, find its median and mode.

Ans : In a symmetrical distribution the mean, median and mode are all equal because the data is evenly distributed around the center. So, its median and mode are 40.75.

(c) Describe the concept of sampling in brief.

Ans : The methods for selecting sample from a population is known in sampling.

(d) Ascertain the range of – 3, 0, 3, – 5.

Ans : Range = Highest value – Lowest value

$= 3 - (-3) = 3 + 3 = 6.$

(e) What are the components of time series?

Ans : The components of time series are (i) Trend, (ii) Seasonal Variation , (iii) Cyclical Variation and (iv) Irregular variation.

(f) Illustrate the extreme values of probability.

Ans : The extreme values of probability 1.

(g) What is null hypothesis?

Ans : The hypothesis about a population parameter we wish to test is called a null hypothesis.

(h) Mention the name of the scholar who has devised the stem and leaf display.

Ans : The stem and leaf display was devised by the American statistician John Wilder Turkey.

(i) Deduce the formula of inter-quartile range.

Ans : The formula of inter-quartile range is $Q.D = Q_3 - Q_1$.

(j) Which index number is ideal?

Ans : Fisher index number is ideal.

(k) Give a pen-picture of scatter diagram.

Ans : Let,

Capital (in lac Tk.) : X	5	10	15	20	25
Profit (in lac Tk.) : Y	3	4	8	12	18

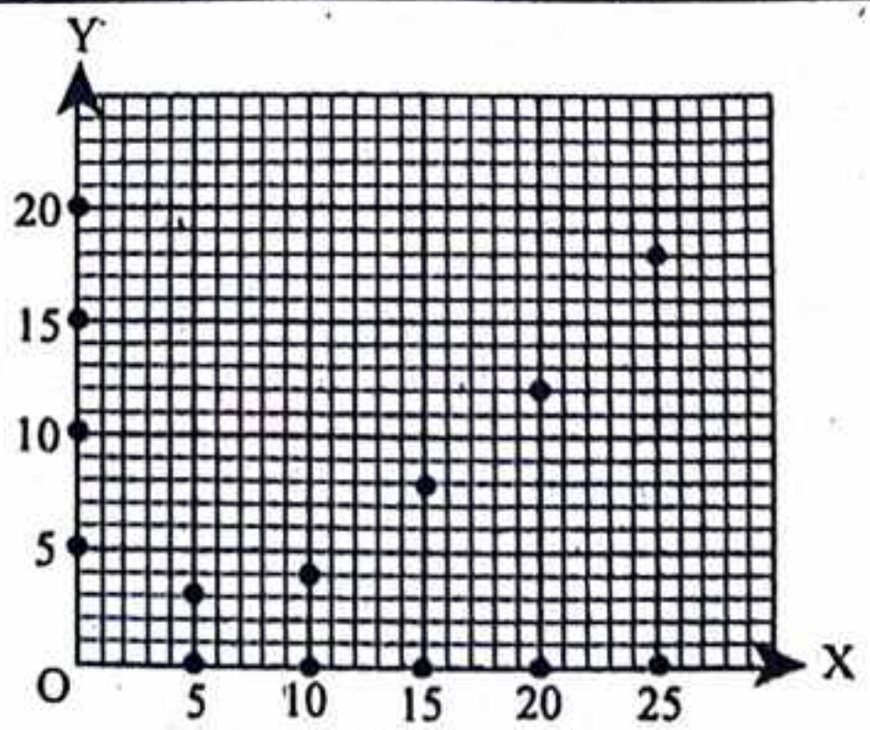


Figure : Scatter diagram

(l) Enunciate the parameters of Poisson distribution.

Ans : The parameters of Poisson distribution are m and λ.

Group-B

[Answer any **five** questions.] Mark— 4 × 5 = 20

2 Show the differences between primary and secondary data.

[Ans পরিসংখ্যান ইন্ট্রোডাকশন (২য় খণ্ড) বইয়ের **Page No. 6 Ques. No 1.12]**

3. Discuss the types of variables. *[Ans : Page No. 9 Ques. No 1.23]*

4. Construct a Stem and Leaf Display from the following data : **Solution :** *Page No-44; Problem No. 20]*

10	8	7	5	10	13	25	21	30	16	19	40
30	50	28	29	32	33	20	26	37	39	24	42

5. The following set of data 13, 42, x, 7, 51, 69, 28, 33, 14, 8 has a median of 29; find out x.
Solution : *Page No-68; Problem No-04.*

6. On a certain examination, the average grade of all students in section A is 71.2. If the average of both classes combined is 70; what is the ratio of the number of students in section A to the number in section B?

7. Regression co-efficient of x on y is 3.5, regression co-efficient of y on x is 0.8. Is it true? Show the causes.
Solution : *Page No-293; Problem No-35.*

8. A bag contains 6 red and 8 green balls. If one ball is drawn at random, then what is the probability of the ball being green? **Solution :** *Page No-176; Problem No-05.*

9. Calculate 4-yearly moving average of the students of 3rd year BBA class;

Year	2016	2017	2018	2019	2020	2021	2022
No. of students	45	60	50	70	65	80	90

Solution : *Page No-312; Problem No-06.*

Group-C

[Answer any **five** questions.] Marks — 10 × 5 = 50

10. Draw an ogive from the following table and calculate the median from ogive.

Wages per hour	10 – 15	15 – 20	20 – 25	25 – 30	30 – 35	35 – 40
No. of workers	8	11	12	16	12	7

Solution : *Page No. 35 Problem No. 13*

11. Find out the missing frequencies from the incomplete table. When the median is 56. Total frequencies is 230.

Class interval	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80	80 – 90
Frequency	12	30	?	65	?	26	18

Solution : *Page No-80; Problem No-12.*

12. Compare the efficiency of two players from the given score : **Solution :** *Page No-123; Problem No-11.*

Cricketer-A	98	17	0	29	70	45	3
Cricketer-B	26	32	53	35	26	17	52

13. The data relating to sales and profit of the five shops are given :

Sales (In lakh Tk.)	10	20	30	40	50
Profit (In lakh Tk.)	3	4	8	12	17

You are required to :

(i) Draw a scatter diagram;

(ii) Compute the co-efficient of correlation and interpret its value.

Solution : *Page No-261; Problem No-13.*

14. Fit a trend line by using least square method from the following data and estimate the sales for the year 2025.

Year	2017	2018	2019	2020	2021	2022	2023	2024
Sales (In lakh Tk.)	35	38	62	71	70	62	89	97

Solution : *Page No-318; Problem No-13.*

15. A researcher collected the following information for two variables x and y;

No. of pairs, n = 20, r = 0.5, $\bar{x}$ = 15, $\bar{y}$ = 20, σ_x = 4, σ_y = 5. Later on, it was found that one pair of value has been wrongly taken as : x = 16, y = 30; where as the correct values were : x = 26, y = 35. Find the correct value of r.

Solution : *Page No-253; Problem No-06.*

16. A simple random sample of size 36 is drawn from a finite population consisting of 101 units. The population's standard deviation is 12.60. Find the standard error of sample mean when the sample is drawn :

(a) With replacement;

(b) Without replacement. **Solution :** *Page No-235; Problem No-23.*

17. Determine Fisher's index number by using the following data and verify the time reversal test and factor reversal test : **Solution :** *Page No-343; Problem No-07.*

Commodity	2023		2024	
	Price	Quantity	Price	Quantity
A	8	10	10	12
B	8	14	5	17
C	5	20	8	18
D	17	6	15	৬

National University

[BBA Hon's 3rd Year; Exam-2024 (Held on- 29/01/2026)]

(Department of Management)

Subject Code : 232603

Subject Name : Business Statistics (In English)

Time : 4 hours; **Full marks :** 80

[**N.B.** Different parts of each question must be written consecutively.]

Part-A

1. Answer any **ten** questions : $1 \times 10 = 10$

(a) What is an independent variable?

Ans : The known variable in regression analysis is called independent variables.

(b) What is secondary data?

Ans : When an organization on investigation uses the data that has already been collected and used by others, such data are called secondary data.

(c) What is Struges rule?

Ans : Struges rule is a statistical rule used to determine the number of class intervals when constructing a frequency distribution.

(d) What is standard deviation?

Ans : The standard deviation is the positive square root of the mean of the squared deviations from their mean of a set of value.

(e) Elaborate ANOVA.

Ans : ANOVA = Analysis of Variance.

(f) What is correlation?

Ans : The term correlation mean the combination of two interrelated on dependable variance. Practically the interrelation between two or more variables are called correlation.

(g) What is TRT?

Ans : Time reversal test is a test to determine whether a given method will work both ways in time, forward and backward.

(h) What is cluster sampling?

Ans : Cluster sampling is a method of random sampling in which the population is divided into groups.

(i) What is T-test?

Ans : T-test is a statistical hypothesis test used to determine whether there is a significant difference between the means of one or two groups when the sample size is small or less than 30 and the population standard deviation is unknown.

(j) Define null hypothesis.

Ans : The hypothesis about a population parameter that we wish to test is called the null hypothesis.

(k) What is time series analysis?

Ans : When data are arranged in against of time, it is called the time series.

(l) What is skewness?

Ans : Skewness is the degree of asymmetry or departure from symmetry in a distribution.

Part-B

[Answer any **five** questions.] Marks : 4 × 5 = 20

2. Discuss the differences between statistics and statistic with example.

[Ans : বিস্তারিত উত্তর (২য় খণ্ড) বইয়ের Page No. 562 Ques. NO- 2]

3. What are the methods of collecting secondary data? *[Ans : Page No. 7 Ques. No 1.14]*

4. From the following data draw a histogram :

Class interval	20 – 25	25 – 30	30 – 35	35 – 40	40 – 45
Frequency	8	12	15	10	8

Solution : *Page No- 562 Question No-04*

5. The arithmetic mean of two observations is 15 and their geometric mean is 9. Find out their harmonic mean.

Solution : *Page No- 563 Question No-05*

6. From the following data calculate the standard deviation :

8, 10, 11, 9, 10, 11, 15, 10

Solution : *Page No- 563 Question No-06*

7. Calculate correlation between price and demand from the following data :

Price (Tk.)	3	5	6	8	9
Demand (Unit)	3	4	5	7	8

Solution : *Page No- 564 Question No-07*

8. A coin is tossed three times. What is the probability of getting a head in all the three tosses?

Solution : *Page No- 565 Question No-08*

9. Find the coefficient of skewness from the given data :

Size	5	6	7	8	9	10
Frequency	14	16	17	21	17	15

Solution : *Page No- 565 Question No-09*

Part-C

[Answer any **five** questions.] Marks : 10 × 5 = 50

10. (a) Discuss the importance of secondary data in statistics. *[Ans : Page No. 566 Ques. NO- 10 (a)*

(b) What are the differences between qualitative and quantitative data? *[Ans : Page No. 566 Ques. NO- 10 (b)]*

11. Compute mean, median and mode from the following table. Comment about the nature of the distribution :

Height (inches)	60 – 62	62 – 64	64 – 66	66 – 68	68 – 70	70 – 72
No. of students	15	18	26	21	11	9

Solution : *Page No- 567 ; Question No-11*

12. The arithmetic mean and standard deviation of 100 items are found to be 30 and 10. But while calculating them two items are wrongly taken as 31 and 71 instead of 13 and 17. Find the correct mean and standard deviation.

Solution : *Page No- 568 ; Question No-12*

13. Calculate skewness and kurtosis from the following information and comment on the results :

Population in family	2	3	4	5	6	7
Number of family	250	310	400	200	120	70

Note : সমাধানকৃত অংকের সাথে 2024 সালের 13 নং অংকের পার্থক্য 7তম তথ্যে 75 এর স্থলে 70 দেওয়া আছে। তাই 2024 সালের অংকটি সমাধানের সময় 7তম তথ্যে 75 এর স্থলে 70 ধরতে হবে।

Solution : *Page No-162; Problem No-44.*

14. From the following information calculate two regression equations and draw the regression line :

Advertisement (Tk. in lac)	40	45	48	50	55	60	65
Sales (Tk. in crore)	12	14	15	17	18	20	21

Solution : *Page No- 569 ; Question No-14*

15. In a simple random sampling of 600 men taken from a village in which 400 men are found to be farmers. In another simple random sampling of 900 men taken from another village in which 450 men are found farmers. Do the data indicate that there is a significant difference in the occupation of men in the two villages?

Solution : *Page No-208; Problem No-03.*

16. The productions of a rice mill are given below :

Year	2018	2019	2020	2021	2022	2023	2024
Production (In thousand tons)	72	90	98	96	99	102	110

(i) Fit a straight line by the least square method and tabulated the trend value;

(ii) What is the monthly increase or decrease in the production of rice?

Solution : *Page No- 572 ; Question No-16*

17. From the given data construct index number of prices for 2024 with 2021 as base. By using–

(i) Laspeyre's method

(ii) Paasche's method and

(iii) Fisher's method :

Commodity	2021		2024	
	Price per unit	Expenditure (Tk.)	Price per unit	Expenditure (Tk.)
A	2	10	4	16
B	3	12	6	18
C	1	8	2	14
D	4	20	8	32

Solution : *Page No-337; Problem No-03.*